DIGITAL DOCUMENT MANAGEMENT AND APPLICATION SYSTEM FOR RESIDENTS OF BARANGAY SAMPIRUHAN

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In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology
Specialized In Service Management Program

CLAIRE MARIE A. ENRIQUEZ
MICHAELA D. STA.CLARA
MEKHAILA S. EUSEBIO

JONALYN JOY B. LABAYNE

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APPROVAL SHEET

The Capstone Project entitled **"Digital Document Management and Application System for Residents of Barangay Sampiruhan"** prepared and submitted by **Claire Marie A. Enriquez, Mekhaila S. Eusebio and Michaela D. Sta. Clara** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

JONALYN JOY B. LABAYNE
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

ROWAN N. ELOMINA
Member

CAROLINA R. JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, Ed. D.
Chairperson, Research and Development Services

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

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DEDICATION

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C.M.A.E.

M.D.S.C.

M.S.E.

ABSTRACT

This study introduces a web-based Digital Document Management and Application System designed to make document submission, processing, and tracking more efficient for the residents of Barangay Sampiruhan. The system enables residents to submit requests online while allowing barangay officials to manage and monitor applications with ease and accuracy. Built using Laravel and supported by Geographical Information System (GIS) mapping, the system ensures data is securely organized and accessible, minimizing manual errors and delays. Findings indicate that the platform improves communication between residents and officials, enhances transparency, and streamlines administrative workflows. By adopting this digital approach, Barangay Sampiruhan can provide faster, more reliable public service, while fostering a more connected and responsive community. In addition to these core benefits, the study highlights how the system encourages greater community participation by making essential services more accessible to residents, especially those who face challenges visiting the barangay office in person. The digital platform also provides a centralized database where documents can be securely stored, retrieved, and managed, ensuring accuracy and consistency in record-keeping. Overall, the system demonstrates the potential of digital innovation in strengthening local governance and improving service delivery, serving as a model for future technology-driven initiatives in similar communities.

Keywords: *Digital Document Management, Web Application, Laravel, GIS, Public Service, Barangay Residents*

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DEFINITION OF TERMS

Administrator	An authorized barangay official or staff member who manages the system, oversees user accounts, approves or rejects applications, and ensures the accuracy and security of stored data.
Application system	A web-based platform that allows residents to submit and process requests for barangay documents digitally.
Barangay Sampiruhan	The local government unit in Calamba City, Laguna, which serves as the project's area of implementation and the primary user community of the system.
Digital Document Management System (DDMS)	A software application designed to store, manage, and track electronic documents, replacing traditional paper-based processes with digital workflows.
Digitization	The process of converting manual, paper-based applications and records into electronic format for faster, more efficient, and paperless transactions.
Document / permit / clearance	Official barangay-issued records such as Business Permits, Barangay Clearances, Certificates of Indigency, and Permit Renewals, generated and stored digitally within the system.
Laravel	A PHP-based web framework used for building modern, secure, and efficient web applications such as the proposed barangay system.

Processing time	The period from the submission of a resident's application until the approval and availability of the final digital document.
Residents	Individuals residing within Barangay Sampiruhan who use the system to request and obtain barangay documents online.
User management	A system feature that allows the administrator to add, update, or remove user accounts and manage access permissions within tha platfrom.
Verification / Notification	A feature that confirms user identity and sends updates about the status of applications through email or mobile number to ensure timely communication.

CHAPTER 1

INTRODUCTION

Project Context

Barangay Sampiruhan, located in the City of Calamba, Laguna, has a population of 9,466 according to the 2020 Census and covers approximately 81 hectares, composed of residential, agricultural, and commercial areas. It is historically significant as the site of the Doce de Pebrero Massacre during World War II, commemorated through the Putol na Kandila monument honoring civilian victims.

With urban growth, the demand for efficient, accessible, and transparent public services has increased. The current manual system for processing barangay documents, business permits, and other administrative transactions has become insufficient. Manual procedures often result in long queues, delayed approvals, misplaced records, and increased workload for barangay personnel, affecting service delivery and limiting administrative efficiency.

To address these challenges, this study proposes the development of a Digital Document Management and Application System for Residents of Barangay Sampiruhan. The system provides a user-friendly, web-based platform enabling residents to apply for and track barangay documents, including barangay clearances, certificates of indigency, business permits. In accordance with Republic Act No. 11261, first-time job seekers are entitled to a free Barangay Clearance, along with certain other documents, as part of government support for employment. The system implemented this provision by automatically identifying eligible applicants requesting a Barangay.

Clearance or Certificate of Indigency, ensuring consistent policy implementation and facilitating access to legally required documents. The system also mitigates risks associated with natural disasters, particularly flooding. Physical records are vulnerable to damage or loss, leading to service disruptions. Digitizing records ensures secure data storage, continuous access to information, and improved operational resilience during emergencies.

Furthermore, the increasing population and number of business establishments necessitate a structured and efficient transaction management system. Automating document processing reduces delays, minimizes human error, and improves transparency.

Implementation challenges include technical issues, potential system downtime, limited internet connectivity in some areas, and varying levels of digital literacy among residents. Handling sensitive personal and financial information also requires robust cybersecurity measures to prevent data breaches and maintain public trust. Addressing these concerns through regular maintenance, training programs, and strong security protocols is crucial for successful adoption.

This study demonstrates that the Digital Document Management and Application System for Residents of Barangay Sampiruhan represent a significant advancement toward modern, transparent, and resilient local governance. By integrating existing barangay policies, enhancing administrative efficiency, and safeguarding critical records, the system supports the barangay's continued development while improving public service delivery and strengthening community trust. It also empowers residents by providing

them with greater convenience, accessibility, and confidence in the barangay's services.

Project Objectives

This project aims to design and implement a web-based Barangay Application System for Barangay Sampiruhan. Its goal is to streamline the process of requesting barangay documents, making it faster, easier, and more convenient for both residents and staff. Instead of undergoing the lengthy manual process, residents can submit their applications online and wait for review and approval. Once approved, documents can be printed at any time, giving residents flexibility and minimizing unnecessary delays.

1. To develop a notification system that keeps residents informed about the status of their applications, whether approved or declined, through email, ensuring efficient and clear communication.

2. To create a simple and user-friendly web system that allows residents to submit and track applications for barangay documents, including clearances and certificates, entirely online. The system also provides convenient payment options, either in cash or through GCash. First-time applicants for a Barangay Clearance or Certificate of Indigency are exempt from payment, in accordance with Republic Act No. 11261.

3. To evaluate the system's effectiveness and usability by applying the Technology Acceptance Model (TAM) and ISO 25010 standards, ensuring that the system is both user-friendly and meets high-quality criteria.

Project Purpose

The purpose of this project is to provide Barangay Sampiruhan with a digital solution that simplifies and improves the application process for barangay documents, such as clearances and certifications. By developing a web-based Barangay Application System, the project aims to make the process faster and more efficient by reducing manual tasks and helping barangay staff manage requests effectively.

The system minimized paperwork, streamline record-keeping, and improve accuracy in data entry, storage, and retrieval. Residents can submit applications online at any time, enhancing convenience and accessibility, while automatic notifications keep them informed of their application status.

Additionally, the system provided barangay officials with data insights to monitor trends and improve services. It also promotes digital literacy among staff and residents, fostering a culture of technology use and preparing the barangay for future improvements. Ultimately, this project aims to enhance resident satisfaction and contribute to a more efficient, transparent, and digitally empowered barangay governance.

Scope and Limitation

This study focuses on creating and implementing Digital Document Management and Application System for Residents of Barangay Sampiruhan in Calamba, Laguna. The goal is to replace the current manual process for submitting and managing applications for barangay documents with a more efficient digital solution. The system aims to improve the efficiency, accessibility,

and transparency of document applications within the barangay, providing a more organized way to handle requests.

The system featured a web-based application portal that allows residents to apply for documents online. This eliminates the need for in-person visits to the barangay hall, saving time and reducing inefficiency. The portal was designed to be easy to use, guiding users through each step of the application process to ensure they provide the correct information. Once applications are approved, residents were given the option to print their documents, while still completing the official verification at the barangay to ensure authenticity.

Barangay officials, including the captain, secretary, and staff, had access to an admin dashboard to manage document applications, monitor statuses in real-time, and track progress. The system also included tools to generate reports and analyze trends, helping officials make informed decisions and improve administrative processes. It was accessible on both mobile devices and desktops, allowing residents to apply from their smartphones while officials manage data efficiently on desktop computers.

However, some challenges must be considered. Unreliable or limited internet access may prevent residents from using the system or cause delays. Security and privacy are also important as personal data must be protected from unauthorized access. The system required ongoing technical support to address bugs and updates, ensuring smooth operation. Some residents or officials may be hesitant to use the system due to unfamiliarity or lack of trust, and budget constraints may pose a challenge in developing and maintaining the system.

Conceptual Framework

The conceptual framework of the Digital Document Management and Application System for Residents of Barangay Sampiruhan illustrates the structured flow from identifying existing problems to delivering a functional and efficient digital solution for barangay document processing. The process begins by gathering inputs, including challenges with the current manual system such as delays, errors, and record losses caused by flooding. It also collects stakeholder requirements, regulatory guidelines, development resources, and evaluation standards like the Technology Acceptance Model (TAM) and ISO 25010.

During the Development Phase, the system undergoes requirement analysis, system design, coding, testing, and debugging to ensure reliability and compliance with specifications. The System Operational Functions encompass online registration and submission of applications, payment integration, secure data storage and retrieval, automated notifications, and administrative verification and approval. These functions automate and streamline the entire application process.

The Output Module delivers a fully functional web-based system that enables residents to submit and track their document requests conveniently online. It provides digitized records, printable approved documents, real-time application status updates, and comprehensive reporting tools for barangay officials. Additionally, the system enhances data security and resilience against disasters, reduces waiting times, and increases transparency in barangay service delivery.

Continuous Feedback and Evaluation, based on TAM and ISO standards, form a crucial part of the framework, allowing for ongoing system improvements through user input and performance assessments. This cycle ensures that the system remains user-friendly, efficient, and adaptable to the evolving needs of the barangay community.

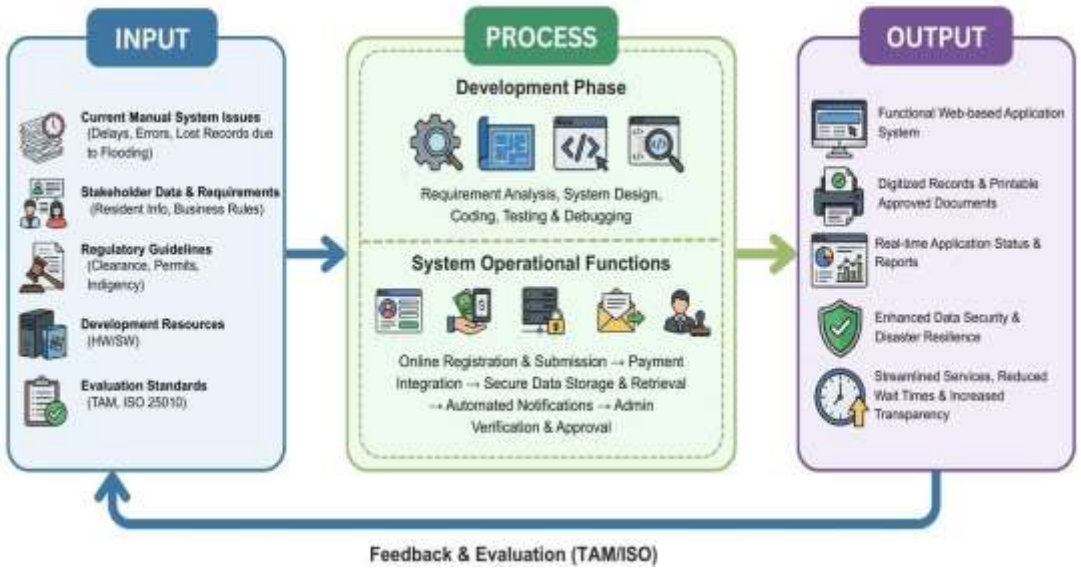


Figure 1. Conceptual Framework

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

The integration of information and communication technology (ICT) in local governance has become essential for promoting transparency, efficiency, and citizen engagement. Many barangays in the Philippines still rely on manual systems, resulting in inefficiency, delays, and errors in processing documents such as clearances and permits (Carpio, 2020). E-governance initiatives digitize these processes, improving accessibility, transaction speed, and service delivery (Bautista, 2015; Siar, 2016).

Digital platforms enhance the efficiency of barangay operations, enabling faster and more accurate service to residents. Automation of business permit processes reduces processing time, improves compliance with regulations, and enhances citizen satisfaction (Gallera & Salvador, 2019; Nuvitek, 2025). AI and automated systems further expedite approvals, manage workloads, and increase operational efficiency while minimizing human errors (Forbes, 2025; Brightly, 2020).

Challenges in adopting digital systems include poor internet connectivity, limited access to hardware, financial constraints, and low digital literacy, particularly in rural areas (National ICT Confederation of the Philippines, 2023; Philippine Statistics Authority, 2022). Addressing these requires infrastructure development, training programs, and stakeholder engagement to ensure successful e-governance implementation (Melendez, 2021).

Digital payment systems complement these efforts by streamlining transactions, increasing speed, transparency, and accountability. QR codebased and mobile payment platforms allow residents to pay for barangay services conveniently, reduce cash handling, and create an audit trail (Manlapig, 2021; Santiago et al., 2020; Gayathri & Shanmugam, 2023). Even small businesses, such as sari-sari stores, benefit from electronic payments, showing increased efficiency, convenience, and income (Technews, 2021; Jaysone, Ildefe, & Benjamin, 2023).

Barangay Management Information Systems (BMIS) and Barangay Management Systems (BMS) provide models for digitizing barangay operations. These systems support record-keeping, SMS notifications, automated permit processing, and other administrative functions, improving operational efficiency and communication between barangay officials and residents (Kazuya, 2015; Mirasol, 2015). The proposed Barangay Sampiruhan system builds on these models to streamline document requests, enhance record management, and offer faster, more transparent services to residents.

Electronic Payment

Electronic payment systems in the Philippines, such as GCash and PayMaya, are widely accessible via mobile phones and do not require a bank account, making them suitable for a largely unbanked population (Zoleta, 2021). Digital payments enable faster processing, reduce manual effort, and improve service delivery in local government operations (Agur, Ari, & Catão, 2020). Effective e-payment strategies integrate technology, business processes, and social networks to enhance convenience, operational efficiency, and customer satisfaction (Jocevski et al., 2020; Swiecka, 2018).

These systems support cashless transactions in barangay services, minimizing physical contact and processing delays.

Operational Efficiency

The adoption of electronic payment systems depends on factors such as technological readiness, perceived security, and ease of use. Businesses that embrace e-payments experience improved operational efficiency, faster transactions, and reduced cash handling, which aligns business practices with customer preferences (Kadjie et al., 2023; Sedigheh et al., 2021). Operational efficiency is further enhanced through automation of repetitive administrative tasks. This reduces staff workload and shortens service delivery time. In barangay operations, improved efficiency leads to better allocation of human resources.

Transaction Speed

Electronic payments provide faster, safer, and more convenient alternatives to cash, enhancing user trust and overall efficiency. Increased transaction speed reduces processing time, minimizes errors, and improves organizational performance for both individual users and businesses (Premchand & Choudry, 2015; Barkhordari et al., 2017). Faster transactions contribute to higher user satisfaction and trust in digital systems. Speed also encourages repeated system use and acceptance. This is essential in barangay services where timely processing is expected.

Barangay Management Information System

BMIS with SMS capability allows residents to access records and updates while enabling barangay officials to manage information efficiently. SMS support requires a broadband connection, though it may not reach all

residents (Kazuya, 2015). The system enhances communication by providing timely notifications regarding document status. It reduces the need for in-person follow-ups. BMIS supports improved coordination between barangay staff and residents. Additionally, BMIS centralizes resident information, making data retrieval faster and more accurate. It minimizes duplication of records and reduces clerical errors. The system also supports informed decision-making by providing real-time data access. Improved information flow enhances transparency in barangay transactions. Overall, BMIS strengthens administrative efficiency and service responsiveness.

Barangay Management System

This system aids barangay administration by allowing client barangays to keep their records up to date and easily accessible for verification, monitoring, and reference. It uses an automated procedure to process barangay clearances and other barangay revenue streams such as levies. Other Barangay-related operations are also available, such as bottle reports and other services. Forms and reports are used to create and print Barangay Permits and Certifications. Standard-based follows section 394 of the Local Government Code, which requires each barangay to keep an up-to-date record of residents for easy identification on the Local Barangay Statistics (Mirasol, 2015).

Barangay Document and Issuance System

The BDIS improves usability, reduces administrative workload, and enhances cloud security through blockchain integration, offering a reliable solution for document processing and record management in local governance

(Balila et al., 2022). High usability ratings indicate strong user acceptance of digital systems. Blockchain integration strengthens data integrity and security. These features demonstrate the potential of advanced technologies in barangay applications. BDIS ensures that digital records are tamper-resistant and traceable. This improves trust in electronically issued barangay documents. The system also supports faster document verification and authentication. Reduced processing time enhances citizen satisfaction. BDIS highlights how emerging technologies can modernize local government services.

Digital Governance and Service Automation

E-government systems improve efficiency, transparency, and accountability by reducing manual processes and enabling real-time communication (Alawneh, Al-Refai, & Batiha, 2013). The success of digital initiatives relies on user-friendly design, stakeholder participation, and training to ensure effective adoption (Heeks, 2018). Automation reduces bureaucratic delays and enhances service consistency. Transparent systems promote public trust in local governance. These principles are central to effective barangay digital transformation. Digital governance also supports data-driven policymaking and planning. Automated services reduce opportunities for corruption and discretionary decision-making. Citizens benefit from improved access to information and services. These systems encourage greater community participation. Overall, service automation strengthens good governance practices at the barangay level.

Comparison of Manual vs. Digital Documentation

Traditional barangay service delivery relies heavily on manual processes such as paper-based record keeping, face-to-face transactions, and physical storage of documents. These methods often result in long processing times, human errors, misplaced records, and limited transparency in transactions (Carpio, 2020). Residents are typically required to visit the barangay office multiple times to submit applications, follow up on requests, and complete payments, which can be time-consuming and inconvenient.

In contrast, digital barangay systems utilize information and communication technology (ICT) to automate document processing, record management, and service delivery. Digital systems enable faster transaction processing, improved data accuracy, and real-time access to records (Bautista, 2015; Siar, 2016). Online application portals allow residents to request documents remotely, reducing physical visits and minimizing delays.

Moreover, digital systems support integrated features such as electronic payments, automated notifications, and centralized databases. Electronic payment platforms improve transaction speed, reduce cash handling risks, and enhance accountability through automated transaction records (Manlapig, 2021). Automated systems also improve transparency by allowing residents to track the status of their requests, increasing trust in barangay services (Gallera & Salvador, 2019).

Synthesis and Relevance of the Study

This study addresses the inefficiencies of the current manual system for processing barangay documents by proposing a Digital Document Management and Application System for Residents of Barangay Sampiruhan. The system is designed to improve accessibility, transparency, and efficiency by enabling online submission of applications, secure data storage, payment integration, real-time status tracking, and automated notifications. By digitizing document management, the system reduces waiting times, minimizes human errors, and enhances the overall experience for both residents and barangay officials.

The project aligns with existing legal frameworks, including Barangay Sampiruhan Ordinance No. 11261 and Republic Act No. 11261, which mandate fee exemptions for first-time applicants requesting barangay clearances or certificates of indigency. By integrating these regulations into the system, the study ensures that services remain equitable and compliant with government policies, providing social support for residents, particularly first-time job seekers.

In addition, the system fosters digital literacy among residents and barangay staff by encouraging the adoption of technology in administrative processes. This promotes a culture of innovation, prepares the community for future technological advancements, and strengthens the capacity of the barangay to deliver efficient and resilient public services.

Literature Matrix

Table 1. Literature Matrix

Author(s) & Year	Title	Purpose	Methodology	Key Findings	Relevance to Capstone
Bautista (2015)	E-Governance in Philippine Local Government Units	To examine the role of e-governance in improving public service delivery in Philippine LGUs	Descriptive analysis of e-governance initiatives in LGUs	E-governance improves efficiency, transparency, and service accessibility	Supports the implementation of a digital barangay system to enhance service delivery
Carpio (2020)	The Role of Barangays in Public Service Delivery	To identify challenges faced by barangays in providing public services	Qualitative analysis of barangay administrative processes	Manual processes cause delays, errors, and inefficiencies in service delivery	Justifies the need for a digital document management and application system
Dela Cruz & Javier (2019)	Barangay Information System	To automate the issuance of barangay clearances and certificates	System development and evaluation using user feedback	Automation reduced processing time and improved record accuracy	Demonstrates how automation can improve barangay document processing
Bautista, Ramos, & Dizon (2020)	Web-Based Barangay Management and Information System	To integrate resident profiling, document requests, and transaction monitoring	Development of a web-based system tested by barangay staff and residents	Improved efficiency and transparency through automation and notifications	Aligns with the proposed system's online application and notification features
Balila et al. (2022)	Barangay Document and Issuance System (BDIS)	To develop a secure digital system for barangay document processing	System evaluation using ISO standards and usability testing	High usability ratings and improved data security through blockchain integration	Supports the use of secure and user-friendly digital document systems
Manlapig (2021)	Integrating Digital Payments in Local Governance	To examine the impact of digital payments on local government operations	Case analysis of LGUs implementing e-payment systems	Improved transaction speed and reduced manual errors	Supports integrating GCash or electronic payments in the proposed system
Gallera & Salvador (2019)	Online Permit Systems: Reducing Delays and Increasing Transparency	To analyze the effectiveness of online permit systems	Comparative study of manual vs. online permit processing	Online systems reduced delays and improved transparency	Reinforces the benefit of online document application and tracking
Kazuya (2015)	Barangay Management Information System with SMS Support	To improve barangay record management and communication	Development of a BMIS with SMS notification features	Enhanced communication but required stable internet	Supports the notification feature of the proposed system
Mirasol (2015)	Barangay Management System	To automate barangay operations and maintain resident records	System development aligned with the Local Government Code	Improved consistency and compliance with record-keeping standards	Provides a functional model for barangay document automation
Alawneh, Al-Refai, & Batiha (2013)	E-Government Systems and Public Service Delivery	To assess the impact of e-government systems on efficiency and transparency	Survey-based evaluation of e-government implementations	Reduced manual processes and increased accountability	Supports the digital governance framework of the proposed system
Heeks (2018)	Information Systems for Public Sector Reform	To identify factors affecting the success of digital government projects	Analytical review of public-sector IT projects	User participation and system usability are critical for adoption	Emphasizes the need for a user-friendly barangay application system

CHAPTER 3

METHODOLOGY

The research project used a quantitative research technique and the Agile software development methodology for Digital Document Management and Application System for Residents of Barangay Sampiruhan Agile was chosen to enable iterative development via sprints, which allows for continual testing and refining of system features. Quantitative data were collected to evaluate system performance. To verify practicality, testing was carried out in both controlled and real-world agricultural situations.

Project Design

The proposed Barangay Document Management and Application System adopt a modular design integrating user interface components, centralized data management, and administrative control to streamline barangay services. Figure X illustrates the system architecture, which consists of the User Module, Application Processing Module, Database Server, and Admin Management Module.

The User Module provides residents with secure access through login and registration functionalities. It offers a range of services including application and renewal of barangay permits, clearance requests, indigency certifications. This module guides users through form submission and fee payment processes, ensuring all necessary information is captured digitally to reduce errors and manual paperwork. Users can also track the status of their applications and download approved documents, increasing transparency and convenience.

The Admin Management Module is accessible through a secure admin login and empowers barangay officials to efficiently handle user management, application verification, and approval workflows. Administrators can review submitted documents, validate payments, approve or reject applications, and generate comprehensive reports for operational insights. This automation of backend processes minimizes manual workload, expedites processing times, and enhances accountability.

By integrating front-end user services with backend administrative functions through a unified database, the system architecture supports improved service delivery, operational efficiency, and transparency in barangay governance. The modular design allows for scalability, enabling future expansions such as additional document types or payment gateways, thus adapting to the evolving needs of local government units.

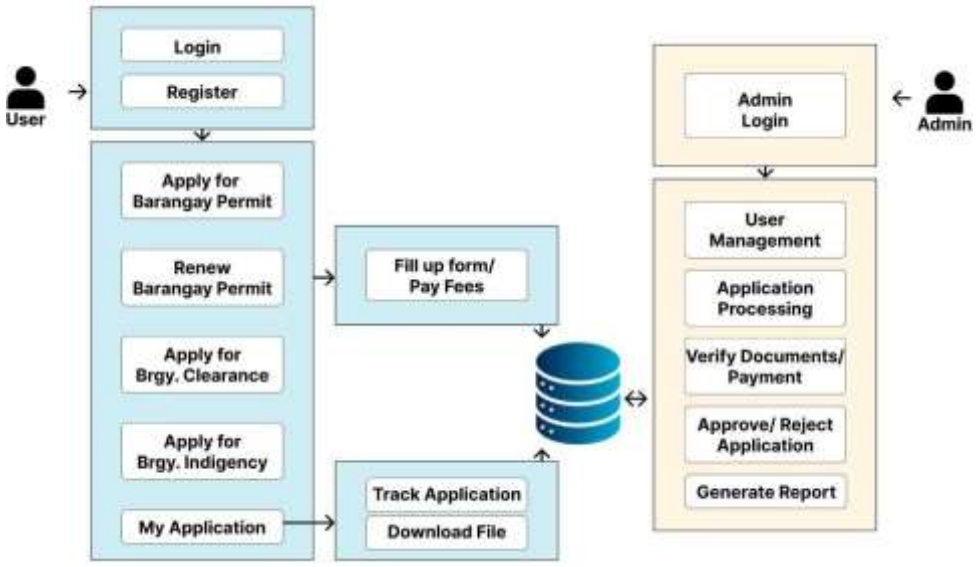


Figure 2. System Architecture

Project Locale

This study was conducted in Barangay Sampiruhan, located in Calamba City, Laguna, a progressive urban barangay situated in the southern part of Luzon, Philippines. Its strategic location near major roads and key establishments makes it an important area for local economic activities. This traditional process requires residents to visit the barangay hall, complete printed forms, and wait for verification and approval to be carried out in person. However, this manual system poses several challenges, including prolonged processing times, the risk of lost or incomplete documents, and difficulty in tracking the status of applications.

Given the barangay's growing population and the increasing number residents in the area, the barangay officials have recognized the need for a digital solution to enhance service delivery. By adopting the Digital Document Management and Application System for Residents of Barangay Sampiruhan, the local government aims to alleviate congestion at the barangay hall, ensure a more accurate and traceable process, and boost service efficiency.



Figure 3. Captain of Barangay Sampiruhan

Block Diagram

The core of the Digital Document Management and Application System for Residents of Barangay Sampiruhan is the web-based application server, which serves as the central processor of all user requests and system operations. The system receives input data from the User Interface, where residents can register, log in, submit applications, and track the status of requested barangay documents such as permits, clearances, certificates of indigency reports.

Once an application is submitted, the data is handled by the Application Submission Module, which validates the information and forwards it to the Database for secure storage. If the application requires payment, the system redirects the user to the Payment Processing Module, where the preferred payment method is selected and verified. After successful verification, the payment status is updated in the database and linked to the corresponding application record.

The system also includes an Admin Module, which allows authorized barangay officials to manage users and process applications. Through this module, administrators can review submitted requests, verify applicant information, approve or reject applications, and update their current status. All administrative actions are recorded in the database to ensure transparency, traceability, and accurate record management.

Additionally, the integrated structure of the system enables real-time monitoring and tracking of application progress for both residents and barangay officials. Residents are able to view updates on their applications through the

user interface, while administrators can efficiently manage workloads and prioritize requests using the admin dashboard. This centralized and automated workflow reduces manual processing, minimizes errors, shortens processing time, and enhances the overall efficiency and reliability of barangay service delivery.

Moreover, the system incorporates access control mechanisms that restrict system functions based on user roles. Residents are granted access only to application-related features, while administrative privileges are limited to authorized barangay personnel. This role-based structure strengthens data security and prevents unauthorized access to sensitive information.

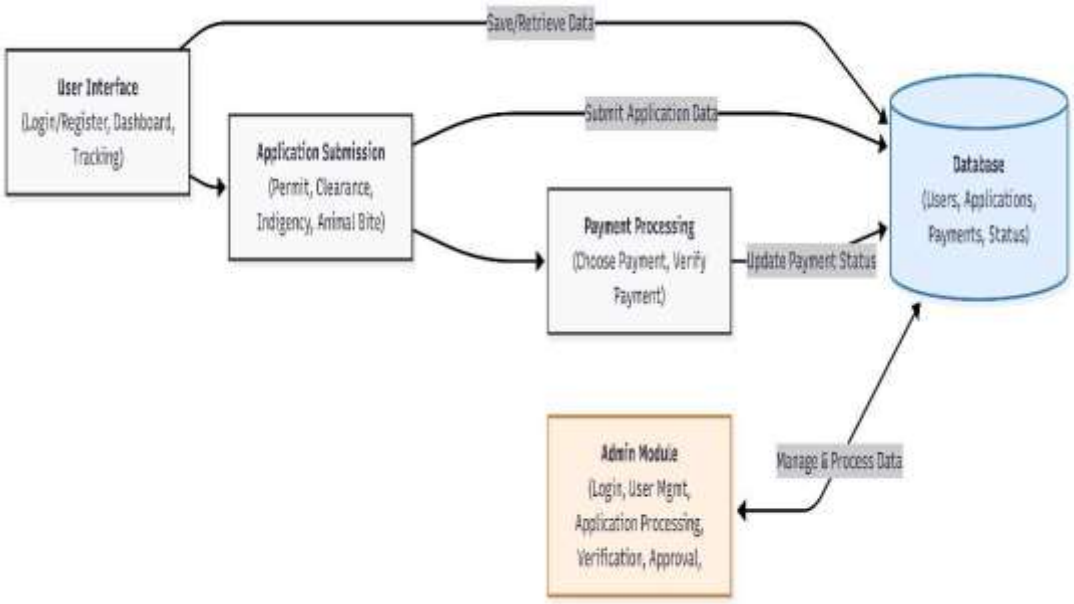


Figure 4. Block Diagram

Software Development

The development of the software component for the project “Digital Document Management and Application System for Residents of Barangay Sampiruhan” involved several key phases, including planning, designing, coding, testing, and refinement. The system was designed to digitize the manual processing of barangay records, allowing residents to submit online requests for documents such as Barangay Clearances, Certificates of Indigency, and Business Permits, while providing officials with a centralized dashboard for managing, verifying, and monitoring these requests.

To guide the development process, the Agile Software Development Methodology was employed. This approach organizes development into iterative cycles, called sprints, which facilitate continuous testing, feedback integration, and progressive refinement of system features. An initial version of the web application was developed and improved through successive iterations, incorporating feedback from residents and barangay officials.

The Agile approach allowed the development team to respond quickly to challenges encountered during implementation, such as ensuring data security, optimizing the interface for mobile users, and integrating specific business rules like the fee exemption for first-time job seekers under Republic Act No. 11261. With each cycle, the software was optimized not only to ensure accurate recordkeeping and faster transaction processing but also to improve transparency, enabling residents to track the status of their requests in real-time.

Materials

This section details the hardware and software tools used in the project, highlighting the components essential for both the physical infrastructure and the development environment. These materials are chosen to meet the project's specific needs for functionality and performance.

Table 2. Hardware Requirements Used for Laptop/Computer

RAM	16GB
HDD/SSD	Samsung SSD 970 EVO Plus
CPU	500GB
GPU	AMD Ryzen 5 5500
CLOCKSPEED	NVIDIA GeForce RTX 2080 3.60Ghz

The hardware specifications for the laptop or computer used in this project are detailed in the table. These include a minimum of 16GB of RAM to ensure sufficient memory for smooth operation. The storage is provided by a Samsung SSD 970 EVO Plus with 500GB, offering fast read and write speeds to accommodate project files and software. The system is powered by an AMD Ryzen 5 5500 processor, which is capable of handling the project's processing requirements efficiently. For graphics, the laptop is equipped with an NVIDIA GeForce RTX 2080, ensuring high-performance visual processing. The system operates at a clock speed of 3.60GHz, which ensures quick processing speeds and optimal performance for complex tasks. These specifications ensure that the computer can handle the demands of the project effectively.

Software

This section describes the software tools and technologies utilized in the development of the Digital Document Management and Application System for

Residents of Barangay Sampiruhan, highlighting the reasons for their selection and their key features. These tools are essential for ensuring an efficient, secure, and reliable system, providing both front-end and back-end functionality. They allow seamless handling of resident requests, data processing, and administrative management, while ensuring compatibility across devices and browsers. The combination of these technologies enables real-time interaction between users and the system, enhances usability, and supports proper data management for improved transparency and service
delivery.

HTML. Is a markup language used to create the structure and content of web pages. It defines the layout and placement of elements such as headings, forms, tables, and links, ensuring proper rendering across web browsers. HTML forms the backbone of the front-end interface, allowing other technologies like CSS and JavaScript to enhance presentation and interactivity.

CSS. Is a styling language used to control the visual appearance of web pages. It ensures proper formatting, spacing, colors, and responsiveness across devices, contributing to a user-friendly and visually appealing design. CSS allows consistent presentation of elements and enhances readability for all users.

JavaScript. Is a client-side scripting language that adds interactivity and dynamic functionality to the system. It enables features such as real-time form validation, modal windows, dropdown menus, and interactive dashboards, enhancing user experience and reducing errors during document submission.

PHP. Is a server-side scripting language used to handle backend processing. It manages tasks such as form input processing, session

management, data validation, and performing CRUD (Create, Read, Update, Delete) operations on the database. PHP ensures smooth communication between the front-end interface and the database.

MySQL. Is a relational database management system used to store and organize system data. It enables efficient data storage, retrieval, updating, and reporting. MySQL ensures that resident information, document requests, and transaction records are securely managed while maintaining data integrity and reliability.

Visual Studio Code. Is a versatile code editor and the primary development environment for the system. It supports multiple programming languages, provides a clean and user-friendly interface, and offers a wide range of extensions that improve productivity, debugging, and code management. Visual Studio Code helps organize project files and streamline development for both front-end and back-end components.

Bootstrap. Is a front-end framework that provides pre-built components and templates for building responsive and consistent web layouts. It simplifies the development of navigation bars, forms, buttons, grids, and modals, allowing faster implementation while maintaining a professional and mobile-friendly design.

Project Development

The system titled “Digital Document Management and Application System for Residents of Barangay Sampiruhan” will be developed using the

Agile Scrum methodology. This iterative and flexible approach allows for continuous feedback and collaboration, ensuring the system meets the specific needs of residents and barangay officials.

The project organized into one- to two-week sprints, with regular stand-up meetings to monitor progress, address issues, and incorporate stakeholder feedback. Key components such as online document submission, real-time request tracking, dashboard management, and secure data handling, were developed and integrated incrementally. This approach enables rapid, adaptive development focused on improving efficiency, transparency, and accessibility in managing barangay documents.



Figure 5. Agile Scrum Methodology

Scope

This research focuses on the design, development, and implementation of a Digital Document Management and Application System for Residents of Barangay Sampiruhan in Calamba, Laguna. The study aims to investigate how a web-based system can improve the efficiency, accessibility, and transparency of barangay document applications, including Barangay Clearances, Certificates of Indigency, and Business Permits.

The system was developed as a web-based application where residents can submit requests online, track the status of their applications in real-time, and access approved documents for printing. The study also examined user interaction with the system on both desktop and mobile devices to determine usability and accessibility.

Barangay officials, including the captain, secretary, and staff, will use an administrative dashboard to manage applications, verify requests, monitor progress, and generate reports for decision-making. The study evaluated how the system supports data security, proper record management, and improved administrative efficiency.

The scope of the research includes system development, testing, deployment, and user evaluation, focusing on core functionalities such as user registration, application submission, real-time tracking, dashboard management, and reporting. The study also considers limitations such as internet connectivity, the need for technical support, and potential user adaptation challenges. Overall, this research seeks to determine the effectiveness of a digital document management system in enhancing

efficiency, transparency, and accessibility within Barangay Sampiruhan, providing evidence-based insights for improving barangay administrative processes.

Design

The system follows a modular, user-focused design to ensure streamlined functionality and ease of use. It integrates several key modules that work together to manage and monitor document applications within Barangay Sampiruhan. The Login Module provides secure authentication for different users, ensuring that only authorized residents and barangay officials can access the system. The User Profile Module allows users to update personal information, modify credentials, and customize their interface preferences. Central to the system is the Dashboard Module, which consolidates and displays real-time analytics of submitted applications, their approval status, and pending tasks for officials. The Application Module enables residents to submit requests for Barangay Clearances, Certificates of Indigency, and Business Permits, guiding users through each step to ensure accurate and complete submissions. The Document Management Module allows officials to review, verify, and securely store uploaded documents. The Notification Module sends automatic alerts to residents regarding application status and notifies officials about pending applications requiring action. Additionally, the Reporting Module organizes historical records and generates reports for analysis, helping officials make data-driven decisions. This modular approach not only ensures clarity and usability but also supports efficient document processing, enhances transparency, and improves overall administrative workflow within the barangay.

Product Backlogs

The Product Backlog for the Digital Document Management and Application System for Residents of Barangay Sampiruhan is a prioritized list of features and functional requirements that define the scope and development plan of the system. The backlog items are organized into major modules for both residents (user side) and barangay staff (admin side), as well as shared features accessible to both groups. For residents, the backlog includes modules that enable secure registration and login, submission of document applications through an online form, uploading of required documents, real-time tracking of application status, notifications for updates, and downloading approved documents. Additional features include viewing past applications, providing feedback, and making secure online payments for applicable fees. Each item ensures residents can interact with the system efficiently, accurately, and conveniently. For barangay staff, the backlog contains modules for secure admin login, reviewing and verifying submitted applications, approving or rejecting requests with comments, generating official digital copies of documents, verifying payments, managing user accounts, generating reports, and tracking system activity through audit logs. These features allow officials to process applications systematically, maintain data integrity, and improve administrative decision-making. Shared system-wide features include mobileresponsive design, secure database management, backup and recovery routines, role-based access control, and activity logging. These elements ensure the system is accessible, secure, and reliable for both residents and officials. This structured backlog serves as the foundation for Sprint Planning and incremental development in the Agile methodology, guiding the team to

build a fully functional, user-friendly, and efficient digital document management system.

Table 3. Product Backlog for User

Story User	Description	User/Role	Priority	Estimated Effort Days
Login Module	As a user, I want secure signin/sign-out so only authorized users can access the system	Resident, Admin	High	1 day
User Profile Module	As a user, I want to view and update my personal information to keep my account accurate	Resident	Medium	1 day
Application Submission	As a resident, I want to submit document requests online so I can avoid in-person visits	Resident	High	1.5 days
Document Upload	As a resident, I want to upload required documents (ID, DTI registration, etc.) to complete applications	Resident	High	1 day
Application Dashboard	As a user, I want a dashboard to track all my applications and their status	Resident	High	1.5 days
Real-Time Status Tracker	As a resident, I want to see the status of my applications (Pending, Approved, Rejected)	Resident	High	1 day
Notification System	As a user, I want to receive notifications about application updates	Resident	High	1 day
Download Permit Certificate	As a resident, I want to download approved documents for official use	Resident	Medium	1 day
Feedback Form	As a user, I want to submit feedback or report technical issues	Resident	Low	0.5 day

The admin stories in the table define the technical functional requirements for barangay staff responsible for managing and processing

residents' document applications. These requirements guide the design and implementation of system modules that support secure, accurate, and efficient administrative operations. The Admin Login module implements secure authentication using encrypted credentials, session management, and rolebased access control to ensure that only authorized personnel can access administrative functions and sensitive resident data. The Application Review and Document Verification modules provide interfaces for filtering, sorting, and validating submitted applications. Uploaded documents are checked against system-defined requirements, and metadata is logged for each verification step. This workflow reduces manual errors and ensures that all application records are accurately processed. The Approve/Reject Applications module integrates decision logic to allow admins to approve or reject requests while attaching comments or reasons. This feature enforces business rules and ensures auditability of all decisions. The Payment Verification module connects to the system's database to confirm transaction statuses, update application records, and flag incomplete or unpaid requests for further action. The Report Generation module executes queries on the database to compile daily, monthly, or yearly summaries of application processing, payment records, and system usage statistics. These reports are generated in real-time or on-demand and are exportable for administrative review and compliance purposes. The System Logging module captures all administrative actions, including logins, application approvals, status changes, and configuration updates. Logs are stored securely in the database with timestamps and user identifiers, supporting auditing, accountability, and troubleshooting.

Together, these modules provide a technically robust system for managing document applications, ensuring data integrity, security, and efficient administrative workflows in Barangay Sampiruhan.

Table 4. Product Backlog for Admin

Story User	Description	User/Role	Priority	Estimated Effort Days
Admin Login	As an admin, I want secure login with encrypted credentials and session management so that only authorized personnel can access administrative functions	Admin	High	1 day
Application Review	As an admin, I want to filter, sort, and review submitted applications so that I can efficiently validate and manage requests	Admin	High	2 days
Document Verification	As an admin, I want to verify uploaded documents and mark them as complete or incomplete so that application records are accurate	Admin	High	1.5 days
Approve/Reject Applications	As an admin, I want to approve or reject applications with comments attached so that decisions are transparent and auditable	Admin	High	1 day
Payment Verification	As an admin, I want to confirm payment statuses and update application records so that only completed transactions are processed	Admin	Medium	1 day
Report Generation	As an admin, I want to generate daily, monthly, or yearly reports of application processing and system usage so that I can monitor performance and make data-driven decisions	Admin	Medium	2 days
System Logging	As an admin, I want the system to log all major actions, including logins, updates, and configuration changes so that auditing, troubleshooting, and accountability are supported	Admin	Medium	1 day
User Management	As an admin, I want to manage resident accounts, reset passwords, and update user information so that user data is secure and up-to-date	Admin	Medium	1.5 days
Barangay Settings	As an admin, I want to update fee rates, application requirements, and schedules so that the system reflects current policies	Admin	Low	1 day

Sprint Planning for User

This sprint plan outlines a five-phase development strategy for the resident interface of the Digital Document Management System. Initial sprints focus on secure account access and online application submission. Subsequent sprints implement real-time application tracking, document downloads with payment integration, and feedback/history review. Together, these sprints ensure the system is secure, efficient, and user-friendly, allowing residents to manage their applications conveniently and transparently.

Table 5. Sprint Planning for User Features

Sprint	Goal	Tasks	Estimated Hours/Day	Expected Outcome
1	Set up secure access for residents	Develop registration and login functionality - Implement password recovery - Add email verification	6–8 hours/day for 2 days	Residents can securely access the system and manage their accounts
2	Enable submission of document applications	Develop online application forms - Implement document upload functionality (ID, DTI registration, etc.) - Validate input data	6–8 hours/day for 3 days	Residents can submit complete applications digitally without in-person visits
3	Provide real-time tracking and dashboards	Develop application dashboard - Implement real-time status tracker (Pending, Approved, Rejected) - Integrate notifications for updates	6–8 hours/day for 3 days	Residents can view and track the status of their applications and receive timely updates
4	Support document retrieval and payment	Implement downloadable permit certificates - Integrate secure online payment gateway - Test payment confirmation and download functionality	6–8 hours/day for 3 days	Residents can download approved documents and pay fees securely through the system
5	Enable feedback and historical data review	Develop feedback form for reporting issues or suggestions - Implement application history log for past submissions	6–8 hours/day for 2 days	Residents can submit feedback and review their past application records for reference

Sprint Planning for Admin

The sprint plan for admin features outlines a four-phase development strategy aimed at providing barangay staff with secure, reliable, and efficient control over the document management system. The first sprint focuses on secure admin access and user management to ensure only authorized personnel can operate the system. The second and third sprints implement application review, document verification, approvals, reporting, and payment verification, enabling staff to process requests systematically and make data-driven decisions. The final sprint ensures system reliability, auditing, and activity logging, supporting accountability and secure administrative operations aligned with the needs of Barangay Sampiruhan.

Table 6. Sprint Planning for Admin Features

Sprint	Goal	Tasks	Estimated Hours/Day	Expected Outcome
1	Set up secure admin login/logout	Implement role-based access control; Develop secure login and logout functionality; Create user management module (add, edit, delete users)	6–8 hours/day for 3 days	Admin can securely access the system and manage user accounts and permissions
2	Configure system approvals	Develop application review module; Implement document verification process; Set up approval and rejection workflow with comments	6–8 hours/day for 3 days	Admin can efficiently process applications while maintaining data integrity
3	Monitor system and generate insights	Develop reporting module; Integrate analytics dashboard; Enable payment verification	6–8 hours/day for 4 days	Admin can generate reports, verify payments, and monitor application trends and status
4	Ensure system reliability and auditing	Implement system logging for all admin activities; Add audit trail tracking; Set up manual override for critical actions	6–8 hours/day for 3 days	Admin can audit activities, troubleshoot issues, and maintain system security

Daily Scrum

The development team holds a daily 1-hour scrum meeting to ensure alignment toward achieving sprint goals for the Digital Document Management and Application System. These sessions allow team members to identify and address obstacles, review progress, and adjust the sprint backlog if necessary. Daily scrums help maintain consistent progress and ensure the system development meets the needs of both residents and barangay staff.

Sprint Execution

During sprint execution, developers work closely with the product owner, including barangay officials and ICT support staff, to define sprint deliverables and estimate workloads. This collaboration ensures development focuses on implementing secure registration, online applications, document uploads, realtime status tracking, notifications, and administrative management features, creating a fully functional, user-friendly system.

Sprint Review Meeting

At the end of each sprint, the development team presents the completed system features, such as user registration, online application submission, document upload, real-time application tracking, and administrative dashboards. This review ensures that each module meets the functional requirements and operational needs of residents and barangay staff.
Stakeholders provide feedback to guide refinements in upcoming sprints.

Sprint Retrospective

After each sprint, the team conducts a retrospective to evaluate work efficiency, collaboration, and the development workflow. Discussions focus on communication, tool usage, and system functionality. Feedback from retrospectives is used to improve processes and ensure smoother development in subsequent sprints.

Project Testing and Evaluation

The system undergoes thorough testing to verify that all modules function correctly and reliably. User-side features like application submission, document uploads, payment processing, and status tracking are tested, while admin-side modules such as application review, verification, reporting, and logging are evaluated for accuracy and security. Testing ensures the system is stable, secure, and user-friendly, ready for deployment to residents and barangay officials.

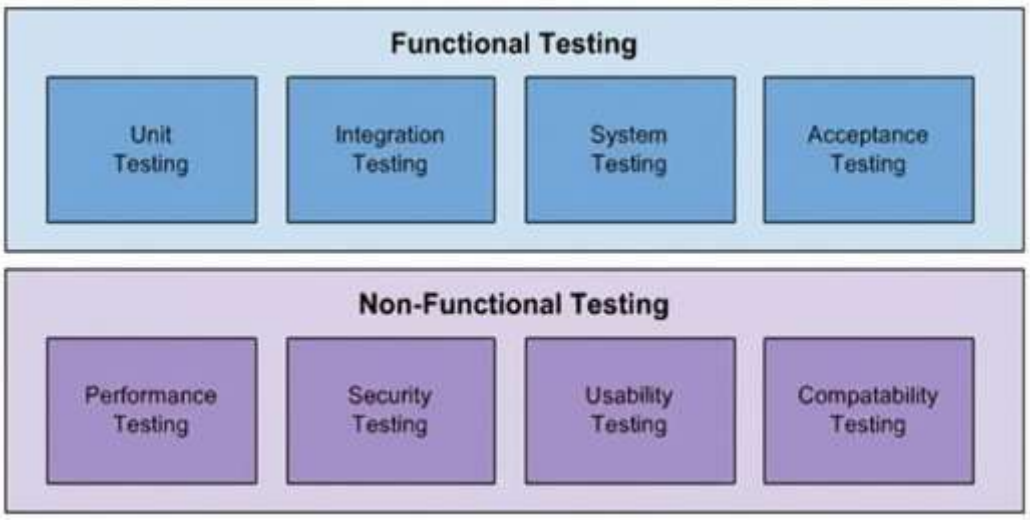


Figure 6. Levels of Testing

Unit Testing

Unit testing was conducted to ensure that each individual component of the system, such as input validation and business name formatting, functioned correctly in isolation. By testing these modules independently, the development team was able to identify and resolve issues at an early stage, ensuring that each feature worked as intended before integration with other components. This process significantly contributed to the overall reliability and stability of the system, reducing errors during full system deployment and providing a solid foundation for subsequent testing phases.

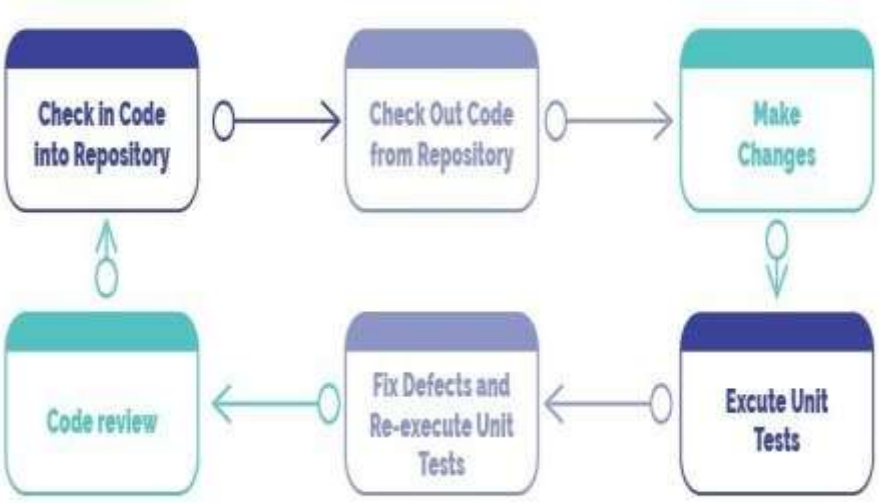


Figure 7. Unit Testing

Integration Testing

Integration testing was performed to verify the interactions between different modules of the Digital Document Management and Application System for Residents of Barangay Sampiruhan. This process ensured that components such as the application form, database, and payment confirmation worked seamlessly together. For example, it validated that submitted applications

correctly updated the database and that payment confirmations reflected in the permit status. By testing these interactions, the development team ensured that modules like user authentication, application processing, and notification systems functioned cohesively, reducing errors in the integrated system.

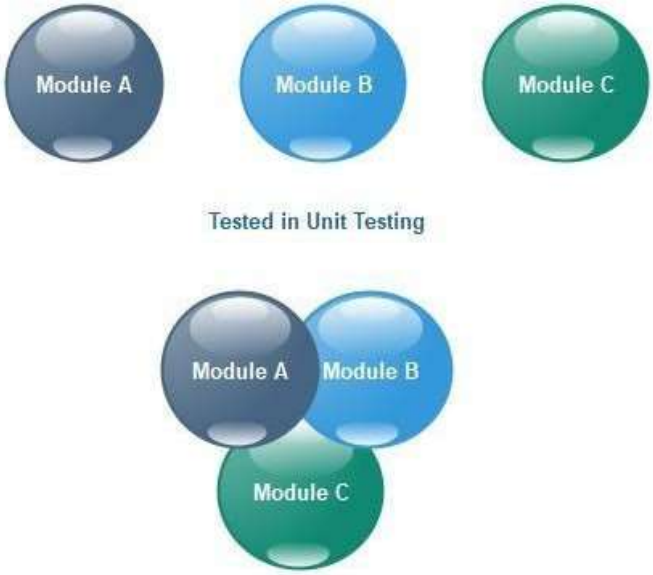


Figure 7. Integration Testing

Acceptance Testing

Acceptance testing was carried out to evaluate whether the Digital Document Management and Application System for Residents of Barangay Sampiruhan met the specific needs of barangay officials and residents. This process involved verifying the correct implementation of key system features such as online document application submission, real-time application status tracking, automated notifications, payment verification, and administrative approval workflows. Final acceptance was granted after the system successfully aligned with the functional requirements, usability expectations,

and user acceptance criteria defined by stakeholders, indicating that the system was ready for deployment in an actual barangay operational environment.

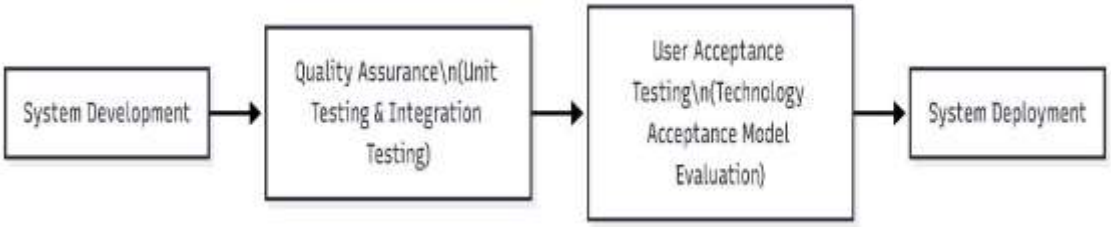


Figure 8. Acceptance Testing

System Flowchart

Figure 9 presents the user flow of the Digital Document Management and Application System, which illustrates how users interact with the system from access to application completion. The process begins by checking whether the user already has an account, where new users are required to register before logging in, while existing users proceed directly to login and access the dashboard. From the dashboard, users can apply for barangay services such as Business Permit, Barangay Clearance, and Barangay Indigency, with options to apply for new permits, renew existing ones, or fill out personal information as required. The system guides users through application validation, payment selection when applicable, and submission of requests. Additional features include profile management, application status tracking, PDF document downloads, and access to barangay contact information and FAQs. Overall, the flowchart demonstrates a structured, efficient, and user centered process designed to streamline barangay service applications and improve accessibility through a centralized digital platform.

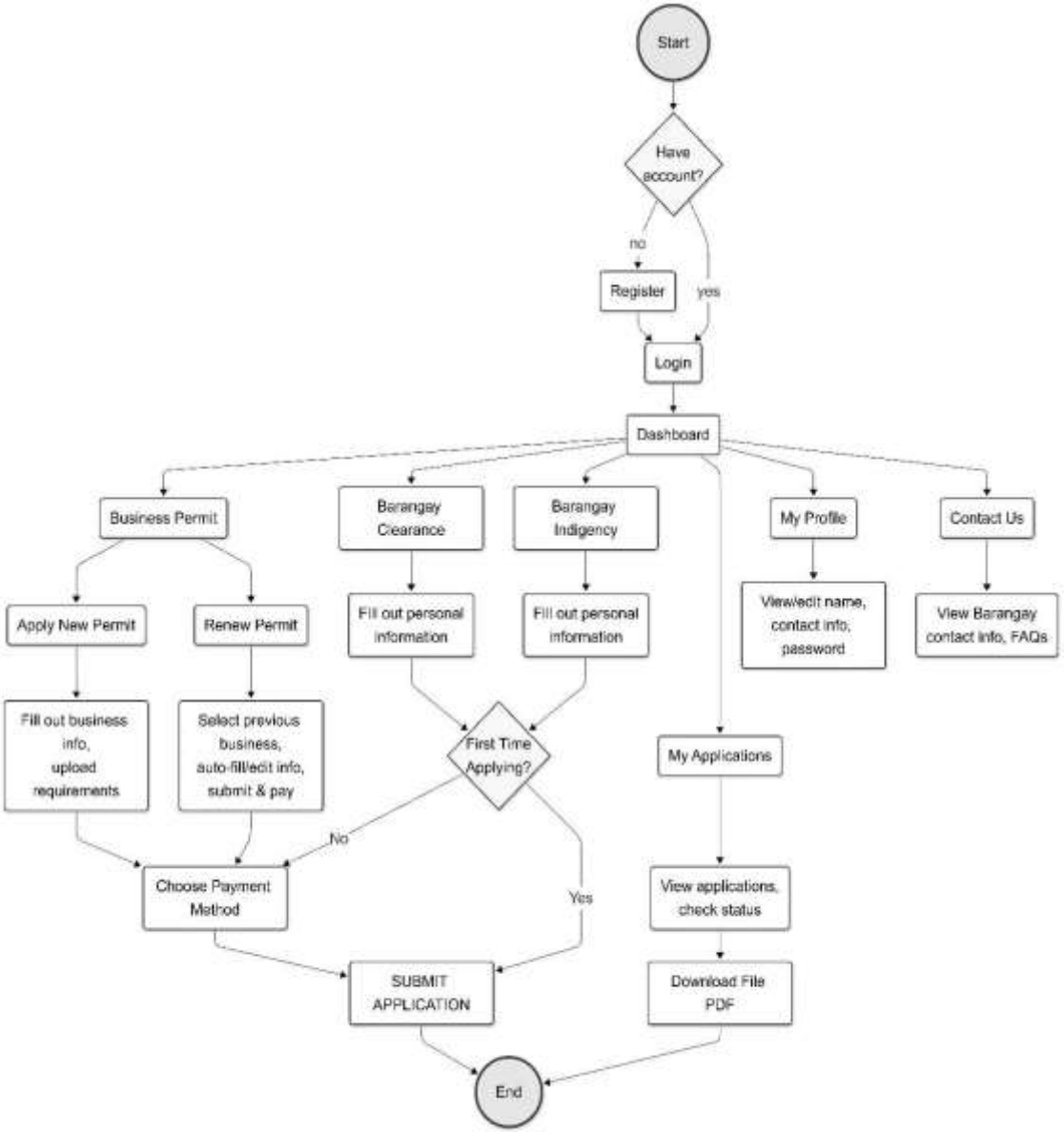


Figure 9. System Flowchart of User

Figure 10 illustrates the admin flowchart of the Digital Document Management and Application System, showing the sequence of tasks performed by the administrator from login to application resolution. The process starts with the admin logging into the system and accessing the admin dashboard, which serves as the control center for system operations. From the dashboard, the admin can manage user accounts by viewing, editing, or deleting records, and process submitted applications by reviewing pending

requests, checking application details, and verifying payments. Based on the evaluation, the admin may approve, reject, or return applications for correction, with corresponding actions such as generating an electronic permit, adding reasons for rejection, or notifying users. The system automatically updates the application status and sends notifications to users, ensuring transparency, accountability, and efficient management of barangay service applications.

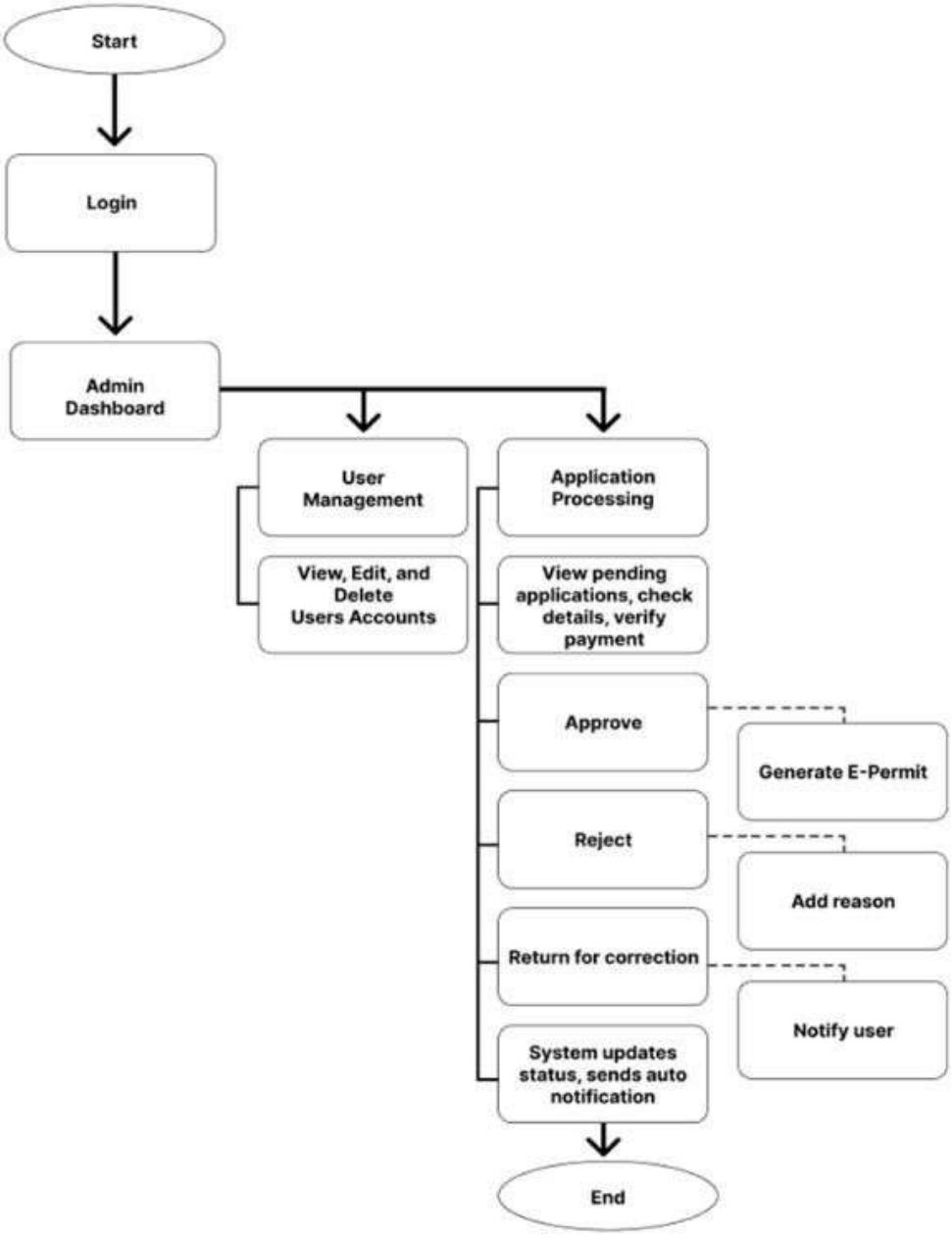


Figure 10. System Flowchart of Admin

Use Case Diagram

Figure 11 illustrates the use case diagram of the Digital Barangay Application System, which defines the functional requirements of the system and the interactions between its primary actors: the user and the administrator. The user is responsible for initiating service requests such as applying for barangay permits, clearances, indigency certificates as well as managing account information, tracking application status, and receiving system notifications. The application process includes mandatory activities such as filling out required information, uploading supporting documents, and completing payment, which are represented as included use cases to ensure completeness and consistency of submitted requests. On the administrative side, the administrator manages user accounts, reviews submitted applications, verifies payments, and performs decision-making actions such as approving or rejecting applications. Approved requests result in the generation of electronic permits, while both approval and rejection actions trigger automated notifications to inform users of the application outcome. The use case diagram also emphasizes system automation by minimizing manual intervention through digital validation and electronic document generation. Furthermore, it supports transparency by allowing users to monitor application progress in real time. Overall, the diagram serves as a blueprint for system development by aligning user needs with administrative processes and system functionalities.

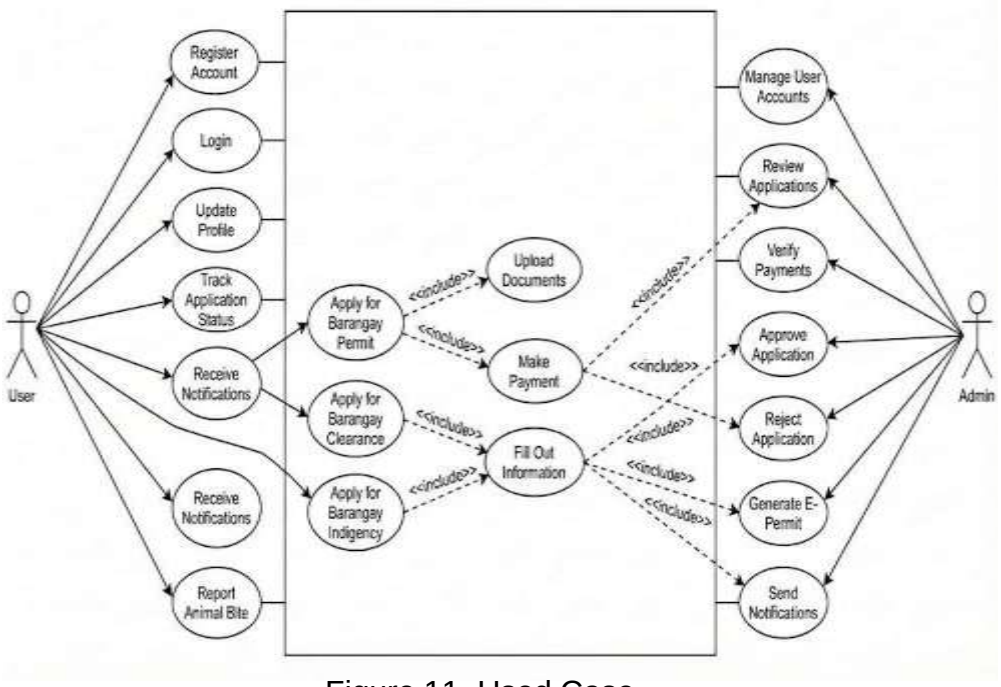


Figure 11. Used Case

Evaluation Procedure

For the Digital Barangay Application System, beta testing was conducted to evaluate the system's overall quality, functionality, and performance. Feedback was gathered from IT specialists using a Likert scale, focusing on key software quality attributes such as usability, efficiency, reliability, and security. The evaluation procedure was guided by the ISO/IEC 25010 quality model, which served as the standard framework for assessing the system.

The assessment emphasized critical quality characteristics including functional suitability, performance efficiency, maintainability, and portability to ensure that the system meets established software engineering standards for digital government services. Results from the evaluation were analyzed to identify areas for improvement, which were subsequently addressed to enhance system performance and user experience. Overall, adherence to the ISO/IEC 25010 model ensured that the proposed system achieved an acceptable level of quality, reliability, and effectiveness for barangay-level application processing and service delivery.



Figure 12. Quality Characteristics of ISO 25010 Standard

Population of the Study

For the Digital Barangay Application System, beta testing was conducted to evaluate the system's overall quality, usability, and performance. Feedback was gathered from three (3) IT expert using a Likert scale questionnaire, focusing on key software quality attributes such as functionality, usability, efficiency, reliability, and security. The evaluation procedure was guided by the ISO/IEC 25010 software quality model, which provides a standardized framework for assessing system quality.

The assessment emphasized critical quality characteristics including functionality, security, maintainability, and portability to ensure that the system meets established software quality standards for digital government services. The results of the evaluation were analyzed to identify areas for improvement, which were used to refine the system and enhance its effectiveness and user experience. Adhering to the ISO/IEC 25010 model ensured that the system delivers reliable, secure, and efficient barangay service transactions.

Table 7. Respondents of the Research Project

CATEGORY OF RESPONDENTS	SAMPLE
IT Expert	3
Barangay Staff	4
Residents	96
Total	**103**

Data Collection Instrument

To evaluate the effectiveness, usability, and reliability of the proposed Digital Barangay Application System, the researcher utilized structured assessment tools designed to gather feedback from the intended system evaluators. The primary objective of the evaluation was to determine how well the system meets the operational needs of barangay transactions, including application processing, information management, and overall user acceptance. The feedback collected served as a basis for identifying areas for improvement and for validating the system's overall performance.

The evaluation was conducted through structured surveys administered to selected respondents composed of three (3) IT experts, four (4) barangay staff, and ninety-six (96) residents. A 4-point Likert scale was used to measure respondents' level of agreement regarding the system's functionality, usability, efficiency, security, and reliability. The use of a 4-point scale eliminated neutral responses, encouraging participants to provide clear and decisive evaluations. The survey questionnaires were distributed in printed form to ensure accurate data collection and to accommodate respondents who may have limited access to digital platforms.

For data analysis, responses were assigned numerical values as follows: 1 – Strongly Disagree, 2 – Disagree, 3 – Agree, and 4 – Strongly Agree. The weighted mean was computed using the formula:

$$\bar{x} = \frac{\sum(f \times x)}{N}$$

where *f* represents the frequency of responses, *x* is the numerical value assigned to each response, and *N* is the total number of respondents. This calculation provided a quantitative measure of the respondents' overall assessment of the system.

To determine the consistency of responses, the standard deviation was computed using the formula:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

where *x* represents each individual response, *bar{x}* is the mean score, and *n* is the total number of respondents. A lower standard deviation indicates consistency in respondents' evaluations, while a higher value suggests varied perceptions of the system.

All collected data were encoded, tabulated, and analyzed using Microsoft Excel to ensure organized, accurate, and reliable processing of survey results.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presented the results and the discussion supporting the methodology in achieving the objectives.

The study developed a notification system that kept residents informed about the status of their applications, whether approved or declined, through email, ensuring efficient and clear communication.

Notification System

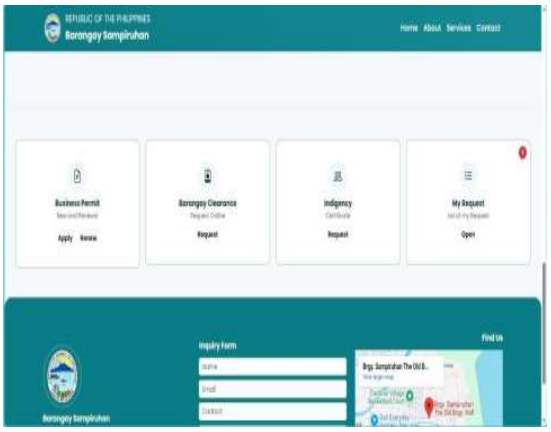
(a) The admin dashboard provides barangay staff with a clear overview of daily operations by displaying real-time counts of permits, clearances, and reports. The monthly trend graph highlights an increase in transactions toward the end of the year, helping administrators anticipate busy periods and manage workloads effectively. Overall, the dashboard simplifies monitoring, improves decision-making, and supports efficient and organized service delivery in Barangay Sampiruhan. *(b)* The system provides applicants with real-time updates on their application status, minimizing the need for in-person follow-ups and improving access to timely information. *(c)* The approved email sample formally informs the applicant of the approval of the Barangay Indigency request and clearly outlines the next steps, including the claiming schedule, office location, and operating hours. By combining in- system notifications with email alerts, the system improves communication, transparency, and timely delivery of accurate application updates.

A simple and user-friendly web-based system was successfully developed, enabling residents to submit and track applications for barangay documents, including Barangay Clearances and Certificates, entirely online.

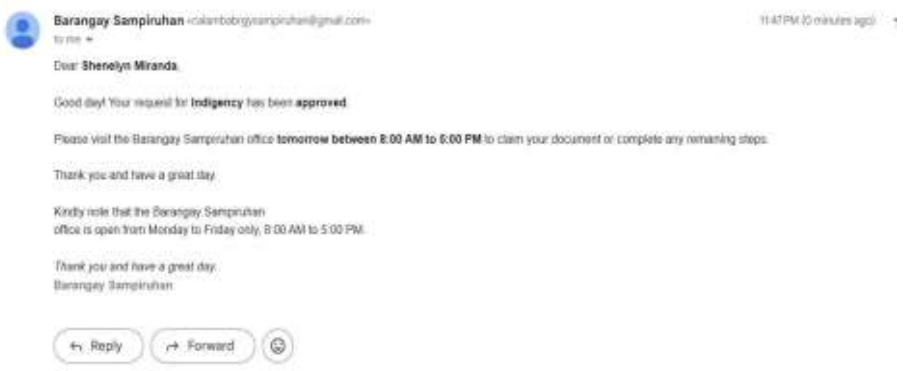
The system effectively supported multiple payment options, including cash and GCash. In compliance with Republic Act No. 11261, first-time applicants for Barangay Clearance and Certificate of Indigency were automatically exempted from payment.



(a)



(b)

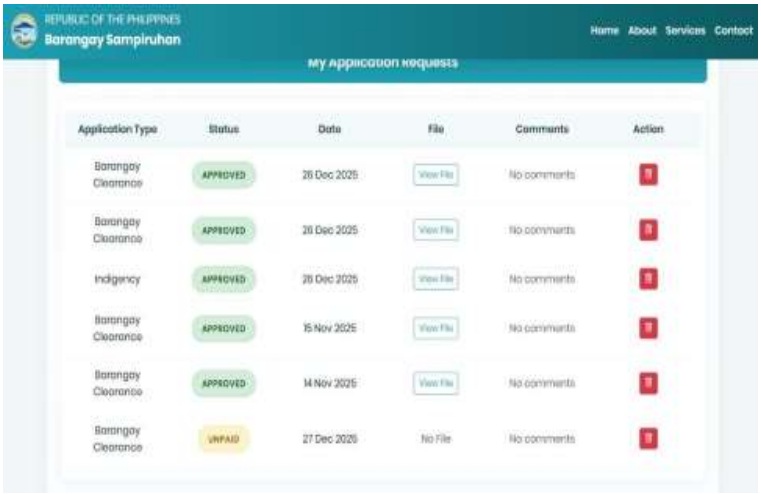


(c)

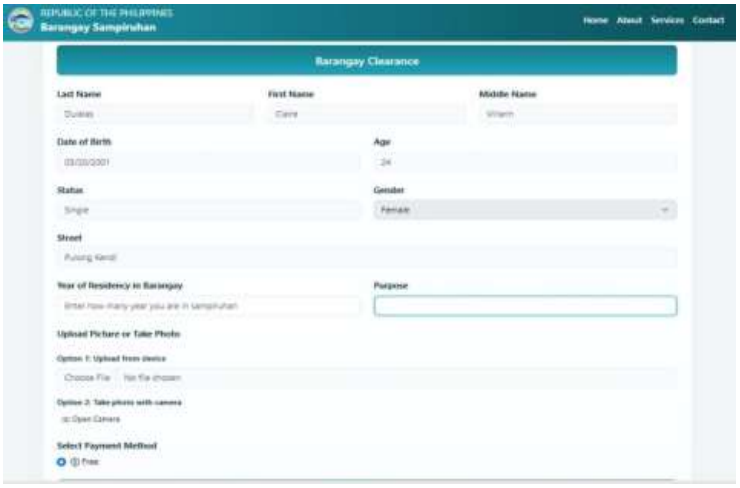
Figure 13. (a) Admin Notification (b) User Notification (c) Email Notification

User Application System

(a) The information is organized into a structured table containing columns for Application Type, Status, Date, and Action. This view utilizes color-coded status badges such as green for "approved" and yellow for "unpaid" to provide immediate visual feedback on the progress of requests, fulfilling the objective of real-time application tracking. *(b)*, the dedicated "Barangay Clearance" request module, is a detailed input view that organizes resident information into logical fields: personal details, year of residency, and purpose of request. The layout includes a modular section for document requirements, allowing users to "Upload from device" or "Open Camera" for real-time photo capture. Notably, this view demonstrates the system's automated compliance logic; under the payment method, it highlights a "Free" status for qualified applicants, ensuring a seamless application of Republic Act No. 11261. *(c)*, the integrated "Payment Method" selection, maintains the clear and modular layout by providing a simplified gateway for financial transactions. It offers residents the choice between traditional "Cash (Pay at Barangay Hall)" and digital "GCash" options, each accompanied by descriptive icons and radio buttons for quick selection.



(a)



(b)



(c)

Figure 14. (a) My Application Requests (b) Barangay Clearance Request (c) Payment Method

Test and Evaluate the Software using ISO 25010.

Table 8 presents the IT expert's evaluation of the system based on the ISO 25010 software quality model. The overall mean score of 4.26 (Excellent) indicates that the system met the expected software quality standards and was highly acceptable. The system received Excellent ratings in Function Suitability (4.33), Usability (4.33), Security (4.42), Maintainability (4.27), Portability (4.58), and Safety (4.60), showing that it is effective, user-friendly, secure, and adaptable. Meanwhile, Performance Efficiency (3.67), Compatibility (4.00), and Reliability (4.11) were rated Very Good, suggesting minor areas for improvement while maintaining dependable performance.Table 8. Results of survey answered by the IT expert

Table 8. Results of survey answered by the IT Expert

Characteristics	Mean	SD	Interpretation
Function Suitability	4.33	0.77	Excellent
Performance Efficiency	3.67	0.58	Very Good
Compatibility	4.00	0.58	Very Good
Usability	4.33	0.75	Excellent
Reliability	4.11	0.53	Very Good
Security	4.42	0.58	Excellent
Maintainability	4.27	0.35	Excellent
Portability	4.58	0.58	Excellent
Safety	4.6	0.58	Excellent
Overall	**4.26**	**0.12**	**Excellent**

Legend: 5.00 - 4.20 Excellent, 4.19 – 3.40 Good, 3.39 – 2.60 Fair, 2.59 – 1.80 Need Improvement, 1.79 – 1.0 poor 2.0

Results of Survey Answered by the Respondents

Table 9 presents the results of the respondents' assessment of the system's Perceived Usefulness. The findings reveal that respondents strongly

agree that the system effectively facilitates document acquisition and service delivery in Barangay Sampiruhan, with a composite mean of 3.34 (Strongly Agree). Specifically, respondents strongly agreed that the system helps in quickly obtaining documents such as clearances and certifications (M = 3.78, SD = ±0.42) and that the integration of digital payment options like GCash provides greater convenience (M = 3.74, SD = ±0.44). Likewise, the notification system through email (M = 3.48, SD = ±0.50) was found to be valuable in informing residents about the status of their applications, ensuring transparency and timely communication.

These results affirm that the system enhances service accessibility and convenience, supporting the shift toward digital governance and automation in local government operations as discussed by Bautista (2015) and Siar (2016). The positive perception of digital payments also aligns with the findings of Manlapig (2021), who emphasized that electronic transactions increase efficiency, transparency, and accountability in public services. Furthermore, the strong agreement regarding system usefulness corresponds with global studies on e-governance (Alawneh et al., 2013; Heeks, 2018), which highlight how automation improves communication, reduces manual errors, and streamlines document processing. Overall, the respondents' feedback demonstrates that the Digital Document Management and Application System for Barangay Sampiruhan is a highly useful innovation that promotes convenience, efficiency, and transparency key indicators of successful e-governance implementation in local communities.

Table 9. Results of Perceived Usefulness

Item	Mean	SD	Interpretation
The system helps in quickly obtaining documents (e.g., Clearance, Certification).	3.78	±0.42	Strongly Agree
The digital payment option (e.g., GCash) offers greater convenience.	3.74	±0.44	Strongly Agree
Does not add to convenience.	2	±0.20	Disagree
The notification system (email) is useful for knowing if the application is approved or declined.	3.48	±0.50	Strongly Agree
The system improves the quality of service provided by Barangay Sampiruhan.	3.68	±0.47	Strongly Agree
Composite	**3.34**	**±0.12**	**Strongly Agree**

Legend: 4.00 - 3.26 Strongly Agree, 3.25 - 2.51 Agree, 2.50 - 1.76 Disagree,1.75 - 1.00 Strongly Disagree

Table 10 presents the results of the respondents' assessment of the Perceived Ease of Use of the developed system. The findings show that the respondents strongly agree that the system is easy to understand and operate, obtaining a composite mean of 3.56 (Strongly Agree). Respondents strongly agreed that the interface design of the web system is easy to understand and use (*M = 3.78, SD = ±0.42*), and that the application process for documents such as clearances is simple and straightforward (*M = 3.70, SD = ±0.46*). They also found that learning the system is quick and effortless (*M = 3.52, SD = ±0.50*), and that selecting payment options, whether cash or GCash, is clear and convenient (*M = 3.36, SD = ±0.48*). Additionally, users strongly agreed that the system can be used without technical assistance (*M = 3.44, SD = ±0.50*). These findings indicate that the system promotes user-friendliness and

accessibility, allowing residents to navigate and complete transactions with minimal guidance. This aligns with the principles of usability and humancomputer interaction highlighted by Davis (1989) in the Technology Acceptance Model (TAM), emphasizing that ease of use enhances user acceptance and satisfaction. Moreover, it supports the goal of e-governance systems to simplify digital processes and encourage citizen participation in online public services.

The results suggest that the Digital Document Management and Application System is user-friendly and efficient for residents of Barangay Sampiruhan."

Table 10. Results of Perceived ease of Use

Item	Mean	SD	Interpretation
The interface (design) of the web system is easy to understand and use.	3.78	±0.42	Strongly Agree
The process for applying for a document (e.g., Clearance) is simple and straightforward.	3.7	±0.46	Strongly Agree
Learning the system is quick and effortless.	3.52	±0.50	Strongly Agree
Selecting and using the payment options (cash/GCash) is clear and easy.	3.36	±0.48	Strongly Agree
The system can be used without technical assistance.	3.44	±0.50	Strongly Agree
Composite	**3.56**	**±0.04**	**Strongly Agree**

Table 11 presents the respondents evaluation of their Behavioral Intention to Use the developed system. The results show that the respondents strongly agree with all statements, with a composite mean of 3.51 (Strongly Agree), indicating a high level of intention to continue using and promoting the system. Respondents strongly agreed that they intend to continue using the system for document applications (*M = 3.54, SD = ±0.50*) and are willing to recommend it to other barangay residents (*M = 3.44, SD = ±0.50*). Likewise, they expressed a preference for the digital system over the traditional manual process (*M = 3.56, SD = ±0.50*), reflecting the system's convenience and efficiency. Additionally, the respondents strongly agreed that they intend to use the system frequently whenever a document is needed (*M = 3.52, SD = ±0.50*) and are likely to utilize the online payment feature through GCash (*M = 3.50, SD = ±0.51*).

These findings imply that the system successfully encourages user adoption and sustained utilization, aligning with the Technology Acceptance Model (TAM), which posits that perceived usefulness and ease of use influence behavioral intention. The results also support studies emphasizing that digital government systems foster citizen engagement and long-term participation when they provide accessible and efficient services (Davis, 1989; Venkatesh & Davis, 2000). Overall, the high behavioral intention ratings indicate that the Digital Document Management and Application System is not only accepted by the users but is also likely to be continuously used and recommended, contributing to the sustainability of digital governance initiatives in Barangay Sampiruhan.

Table 11. Results of Behavioral Intention to Use

Item	Mean	SD	Interpretation
Intention to continue using the system for document applications.	3.54	±0.50	Strongly Agree
Intention to recommend the system to other barangay residents.	3.44	±0.50	Strongly Agree
Preference for this digital system over the traditional process of going to the office.	3.56	±0.50	Strongly Agree
Intention to use the system frequently whenever a document is needed.	3.52	±0.50	Strongly Agree
Intention to utilize the online payment feature (GCash) for all transactions	3.5	±0.51	Strongly Agree
Composite	**3.51**	**0.00**	**Strongly Agree**

Legend: 4.00 - 3.26 Strongly Agree, 3.25 - 2.51 Agree, 2.50 - 1.76 Disagree,1.75 - 1.00 Strongly Disagree

Table 12 presents the respondents' assessment of their Attitude Toward Use of the developed system. The findings reveal that respondents strongly agree with all statements, with a composite mean of 3.68 (Strongly Agree). This indicates that users have a highly positive attitude toward using the Digital Document Management and Application System. Respondents strongly agreed that they enjoy using the system (*M = 3.58, SD = ±0.50*) and that it builds confidence in the barangay's service (*M = 3.84, SD = ±0.37*). They also perceived that the system provides a positive experience in applying for documents (*M = 3.62, SD = ±0.49*) and makes document acquisition more modern and effective (*M = 3.62, SD = ±0.49*). This reflects a high level of user

satisfaction and approval of the system's overall design and functionality. The positive responses suggest that the system successfully enhances the public's perception and trust in barangay digital services. This aligns with the Technology Acceptance Model (TAM), which emphasizes that a positive user attitude increases technology adoption. Moreover, it supports findings from studies on e-governance (Venkatesh & Bala, 2008; Dwivedi et al., 2019), showing that user enjoyment and confidence are crucial factors in the sustained use of government information systems. Overall, the results demonstrate that the developed system not only facilitates efficiency and accessibility but also fosters user satisfaction, confidence, and trust in the barangay's digital transformation efforts.

Table 12. Results of Attitude Toward Use

Item	Mean	SD	Interpretation
Enjoyment while using the system.	3.58	±0.50	Strongly Agree
The system builds confidence in the barangay's service.	3.84	±0.37	Strongly Agree
The digital system provides a positive experience in applying for documents.	3.62	±0.49	Strongly Agree
Feeling that document acquisition more modern effective because of the system.	3.62	±0.49	Strongly Agree
The system builds confidence in the barangay's service.	3.76	±0.43	Strongly Agree
Composite	**3.68**	**±0.06**	**Strongly Agree**

Legend: 4.00 - 3.26 Strongly Agree, 3.25 - 2.51 Agree, 2.50 - 1.76 Disagree,1.75 - 1.00 Strongly Disagree

Acceptance Testing Result

Table 13 presents the results of the acceptance testing for the Digital Document Management and Application System. Users rated all aspects between 3.34 and 3.68, falling under “Strongly Agree.” Perceived Usefulness scored 3.34, showing that users believe the system helps improve productivity and streamline barangay transactions. Perceived Ease of Use received 3.56, indicating that users find it simple and user-friendly. Behavioral Intention to Use scored 3.51, reflecting their willingness to continue using the system, while Attitude Toward Use got the highest score of 3.68, showing a very positive overall impression. With an overall mean of 3.52, the results suggest that the system is well-accepted, meets user expectations, and is ready for implementation in the barangay’s digital operations."

Table 13. Acceptance Testing

ITEM	MEAN	SD	INTERPRETATION
Perceived Usefulness	3.34	±0.12	Strongly Agree
Perceive ease of Use	3.56	±0.04	Strongly Agree
Behavioral Intention to Use	3.51	±0.00	Strongly Agree
Attitude Toward Use	3.68	±0.06	Strongly Agree
Overall	3.52	±0.05	**Strongly Agree**

Legend: 4.00 - 3.26 Strongly Agree, 3.25 - 2.51 Agree, 2.50 - 1.76 Disagree,1.75 - 1.00 Strongly Disagree

Results of Testing and Evaluation

The results of the unit testing show that all user-side features of the system are functioning smoothly and as intended. Users were able to register and log in without any difficulty, and the system consistently redirected them to the correct pages. Core features such as the homepage display, profile management, and information sections loaded properly and presented content clearly. Functional services including applying for business permits, requesting clearances and certificates, and reporting animal bites were all accessible and fully operational. The system also performed well in more advanced features, such as digital payment processing, sending notifications for request updates, and generating downloadable PDF documents. Every test case passed successfully, demonstrating that the system provides a reliable, user-friendly, and efficient experience. Overall, the results confirm that the platform is well prepared for real-world use and meets the needs of its users effectively.

Table 14. Unit Testing (User Side)

Tested Case	Expected Output	Actual Output	Process	Remarks
Account Registration	Users can register by providing their full name, contact number, and email.	Users successfully registered and created accounts in the system.	3 seconds	Passed
Log-in Function	Users can log in by entering valid username and password.	The system verified credentials and redirected users to their respective dashboards.	2 seconds	Passed
Homepage	The system displays the Barangay Sampiruhan homepage with navigation bar, banner, and welcome message.	The homepage successfully displayed with complete navigation, banner, and welcome text.	1 second	Passed
Profile Management	Users can update their personal information in their profile.	User profile information successfully updated and saved.	3 seconds	Passed
About, Mission, and Vision Section	The system displays the “About Barangay Sampiruhan” section along with the Mission and Vision statements clearly and completely.	The system successfully displayed the About, Mission, and Vision sections with proper layout and readable content.	1 second	Passed
Apply for Business Permit	The system directs the user to the “Apply/Renew Business Permit” request form.	The system successfully redirected to the Business Permit application page.	2 seconds	Passed
Request Barangay Clearance	The system displays the Barangay Clearance request form.	The system successfully displayed the Barangay Clearance request section.	2 seconds	Passed
Request Indigency Certificate	The system shows the Indigency Certificate request form.	The system successfully displayed the Indigency Certificate request section.	2 seconds	Passed
Report Animal Bite	The system redirects the user to the Animal Bite Report form.	The system successfully redirected to the Animal Bite Report section.	2 seconds	Passed
Digital Payment Integration	The system allows users to pay through digital options (e.g., GCash).	The payment feature processed transactions successfully.	4 seconds	Passed
View My Requests	The system displays a list of all requests submitted by the user.	The system successfully displayed the list of submitted requests.	2 seconds	Passed

The results of the admin-side system testing show that all key functions are working smoothly and efficiently. The Admin Log-in feature operates without issues, allowing quick and secure access to the dashboard. Once logged in, the Admin Dashboard displays accurate and real-time information, making it easy for the admin to monitor system activity. The Manage Requests function also performs well, enabling the admin to approve, decline, and review user requests without any delays. Lastly, the Generate Reports feature successfully produces downloadable PDF reports, showing that the system can generate accurate and organized documents when needed. Overall, the admin modules demonstrate strong performance and reliability.

Table 15. Unit Testing (AdminSide)

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	PROCESS	REMARKS
Admin Log-in	Admin enters credentials to access the admin dashboard.	Admin successfully logged in and accessed the dashboard.	2 seconds	Passed
Admin Dashboard	The system shows total requests, approved and pending documents, and registered users.	Dashboard displayed accurate and real-time data summary.	2 seconds	Passed
Manage Requests	Admin can approve, decline, or view request details.	Admin successfully approved, declined user requests.	3 seconds	Passed
Generate Reports	Admin can generate transaction and service reports in PDF format.	Reports were successfully generated and downloadable.	4 seconds	Passed

The overall testing results show that the system's core modules are functioning effectively and providing a seamless user experience. The User Account and Log-in Module performed well, securely verifying accounts and directing users to their dashboards without issues. The Application Request and Payment Module also operated smoothly, successfully handling document

requests and processing digital payments. Users were consistently kept updated through the Notification and Tracking Module, which delivered real time status updates as soon as the admin took action. On the admin side, the Dashboard and Request Management features proved reliable by accurately displaying all user transactions and immediately reflecting any updates. The Document Generation Module performed as expected, creating official, downloadable PDFs with the correct details. Meanwhile, the Homepage and Navigation Module ensured smooth movement across system sections, with all links directing users to the proper pages. Lastly, the Report and Analytic Module generated accurate summaries and downloadable reports, confirming its effectiveness in supporting administrative tasks. Overall, the system demonstrated strong reliability, accuracy, and efficiency across all modules.

Table 16. Integration Testing

Tested Case	Expected Output	Actual Output	Remarks
User Account and Log-in Module	The system integrates user registration and login functions, allowing residents to securely access their accounts.	The system successfully verified registered accounts and redirected users to their respective dashboards.	Passed
Application Request and Payment Module	The system connects the document request forms with the digital payment option (e.g., GCash).	The system successfully processed document requests and verified payment transactions.	Passed
Notification and Tracking Module	The system automatically sends notifications once the admin approves or declines requests and updates the status in the user dashboard.	The notification and tracking modules worked seamlessly, updating request statuses in real time.	Passed
Admin Dashboard and User Request Management	The admin dashboard displays all user requests, allowing approval, declination, or viewing of details.	The dashboard accurately reflected user transactions and updates were shown immediately after admin action.	Passed
Document Generation and Download Module	The system integrates the approval process with automatic generation of official documents (PDF).	Approved requests successfully generated downloadable documents with correct details.	Passed
Homepage and Navigation Module	All navigation links (Home, About, Mission, Vision, Services) direct users to their corresponding system pages.	All sections and internal links functioned correctly, displaying the right content per module.	Passed
Report and Analytic Module (Admin)	The system integrates approved transactions into reports.	Reports displayed accurate summaries and were downloadable in PDF format.	Passed

CHAPTER 5

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents a consolidated discussion of the study's results derived from the data analysis. It interprets the findings in relation to the research objectives and the identified problems outlined in the earlier chapters. The implications of the results are discussed to highlight the contribution of the study to the intended users and stakeholders. Based on the findings, relevant conclusions are drawn and practical recommendations are proposed to improve the system and guide future researchers in conducting related studies or further system enhancements.

Summary

The Digital Document Management and Application System for Residents of Barangay Sampiruhan aimed to design and implement a webbased platform that improves the efficiency, accessibility, and transparency of barangay document processing. The study addressed the limitations of the traditional manual system, which often resulted in long processing times, misplaced records, and inconvenience for both residents and barangay personnel.

The researchers developed a web-based system that allows residents to submit applications for barangay documents such as clearances, certificates of indigency, and business permits online. The system features secure user authentication, online document submission, real-time application tracking, automated email notifications, and integrated digital payment options. On the administrative side, barangay officials are provided with a centralized

dashboard to review applications, verify documents and payments, approve or reject requests, and generate reports for monitoring and decision-making.

The system was implemented and evaluated in Barangay Sampiruhan, Calamba City, Laguna. Its effectiveness and quality were assessed using the ISO/IEC 25010 software quality standards and the Technology Acceptance Model (TAM). Evaluation results from IT experts, barangay staff, and residents indicated excellent ratings in functionality, usability, security, maintainability, and overall system performance. Respondents strongly agreed that the system is useful, easy to use, and preferable to the traditional manual process. Overall, the findings demonstrated that the Digital Document Management and Application System significantly enhance service delivery by reducing processing time, minimizing manual errors, improving transparency, and increasing resident satisfaction. The system shows strong potential as a sustainable digital solution for barangay-level governance and may serve as a model for other local government units seeking to modernize public service operations.

Conclusion

Development of an Email-Based Notification System for Application Status Updates

This study successfully developed an email-based notification system that keeps residents informed about the status of their barangay document applications. The system automatically notifies applicants whether their requests are approved or declined, ensuring clear, timely, and efficient communication between the barangay and residents. This feature addressed

delays and uncertainty commonly experienced in manual processing and fulfilled the objective of improving communication efficiency.

Development of a Simple and User-Friendly Web-Based Application, Tracking, and Payment System

The study successfully created a simple and user-friendly web-based system that allows residents to submit and track applications for barangay documents, including clearances and certificates, entirely online. The system provided convenient payment options through cash or GCash. In compliance with Republic Act No. 11261, first-time applicants for Barangay Clearance or Certificate of Indigency were automatically exempted from payment. This objective was achieved by improving accessibility, transparency, and convenience in barangay document processing.

Evaluation of System Effectiveness and Usability Using TAM and ISO/IEC 25010 Standards

The effectiveness and usability of the developed system were evaluated using the Technology Acceptance Model (TAM) and ISO/IEC 25010 software quality standards. The evaluation results indicated high levels of perceived usefulness, ease of use, and user acceptance, as well as strong performance in functionality, reliability, security, and maintainability. These findings confirm that the system meets high-quality standards and is acceptable for actual use in barangay operations.

Recommendations

The Digital Document Management and Application System for Residents of Barangay Sampiruhan was able to meet its objectives and function effectively based on the results and evaluations conducted. However, further enhancements and improvements can be made to maximize its accessibility, efficiency, and user satisfaction. The following recommendations are suggested:

1. Develop a mobile application to enable residents to submit requests and monitor applications directly from their smartphones, increasing convenience and system adoption.
2. Implement a real-time chat feature to allow residents to communicate instantly with barangay staff, improving support and streamlining inquiries.
3. Integrate card payment options such as Visa or Mastercard to offer faster, more secure, and convenient payment methods.
4. Use QR code verification for document requests so that residents can present a unique code at the barangay office, allowing staff to quickly access request details and reduce errors.
5. Add SMS notifications to inform residents about application status, approvals, or required actions, ensuring timely communication and minimizing unnecessary follow-ups.

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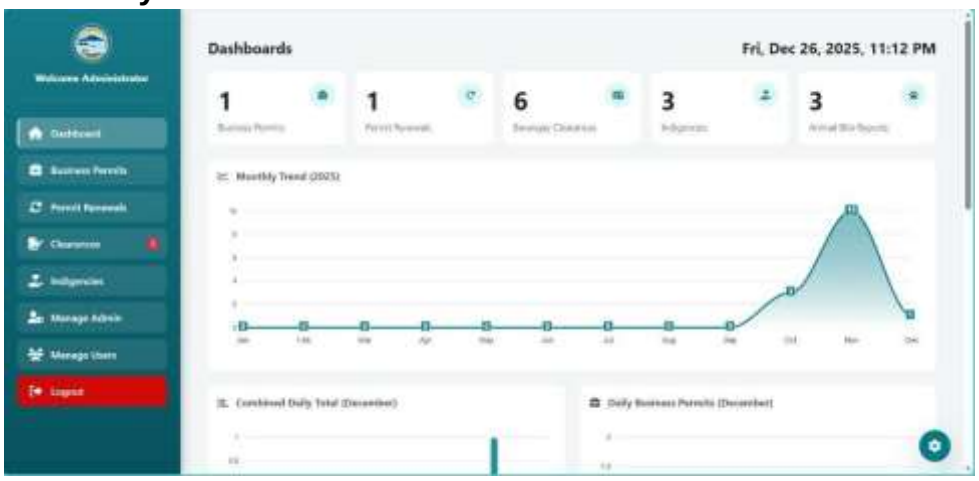
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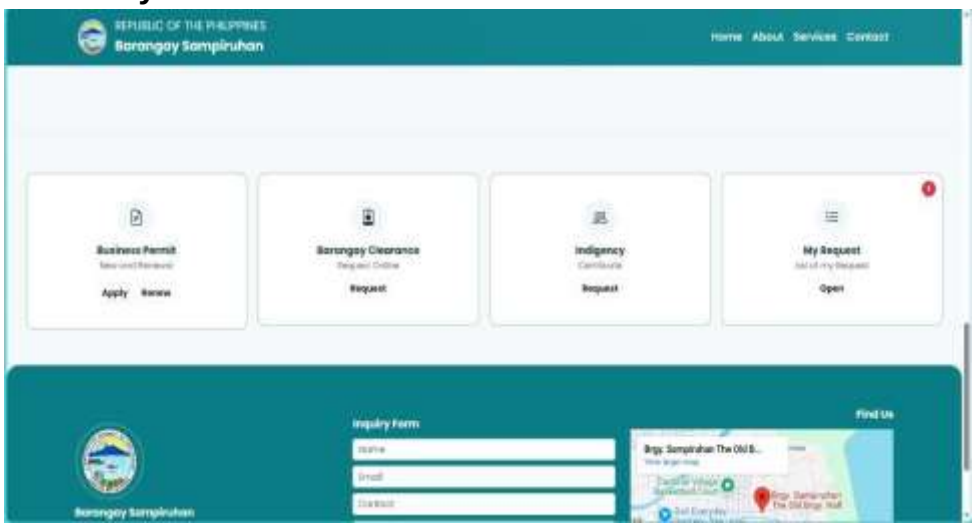
APPENDIX A

Output Screenshot

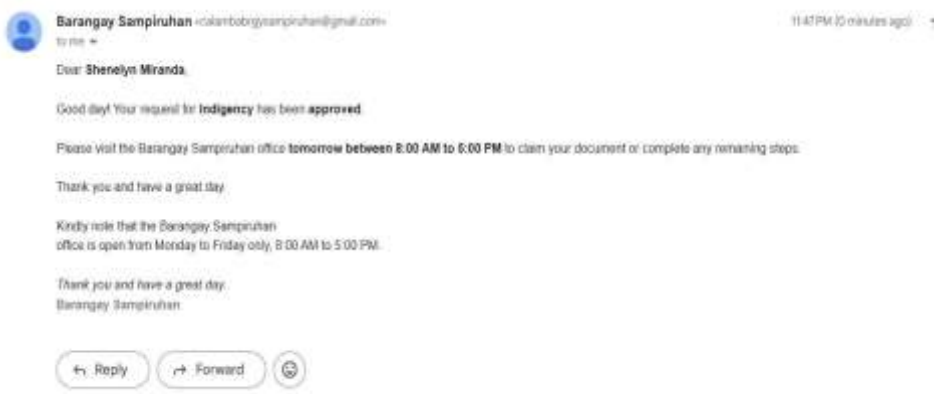
Notification System for Admin



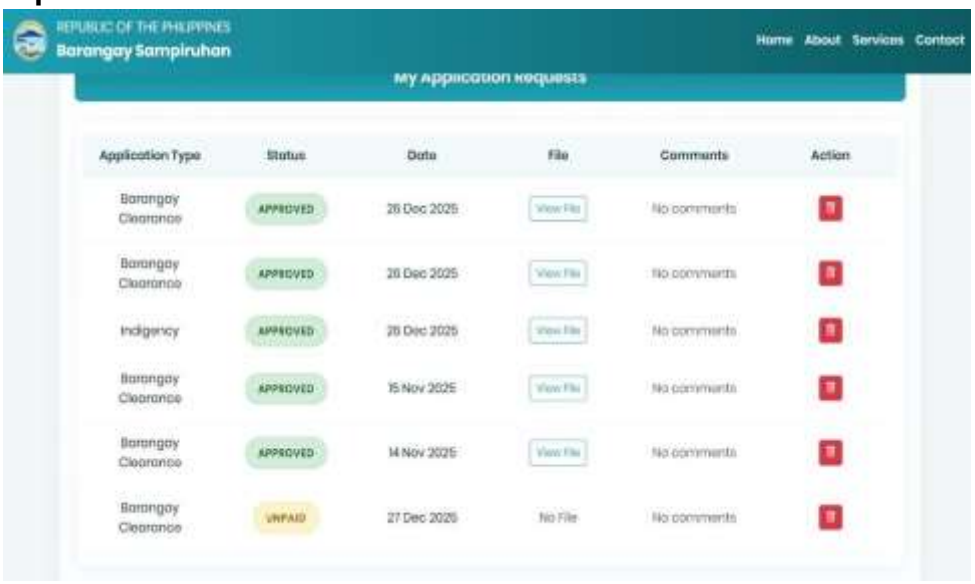
Notification System for User



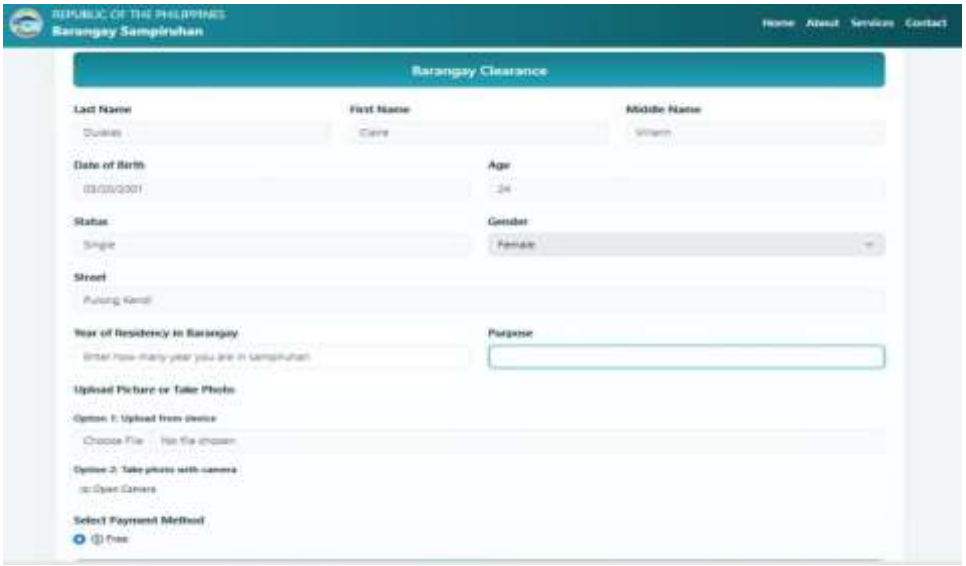
Notification System for User thru Email



My Request



Application



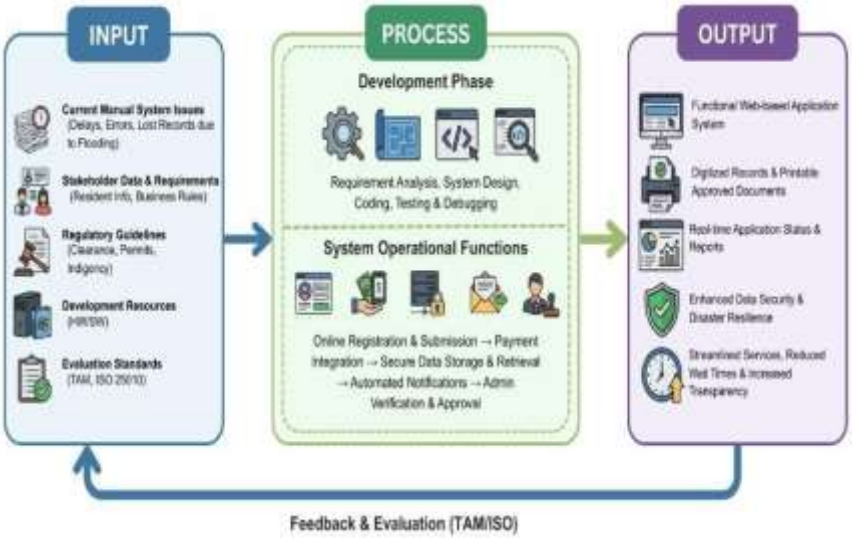
Payment Option



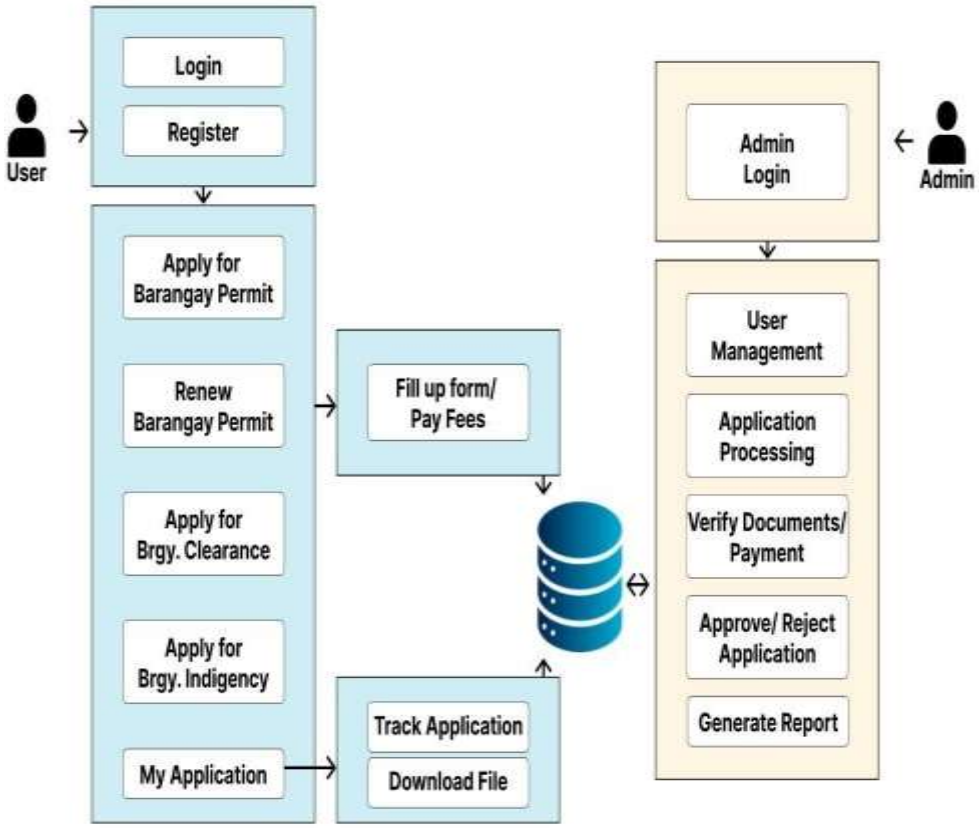
APPENDIX B

Figures

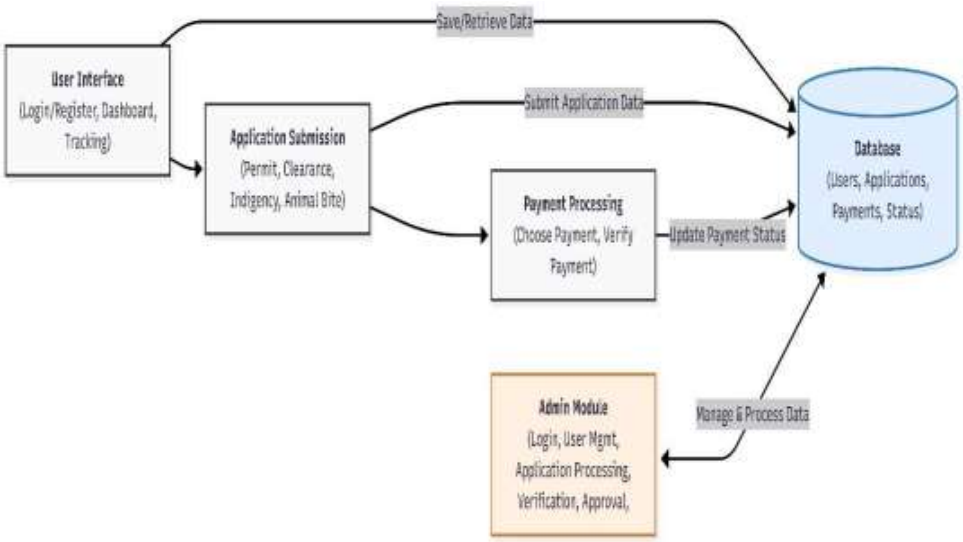
Conceptual Framework



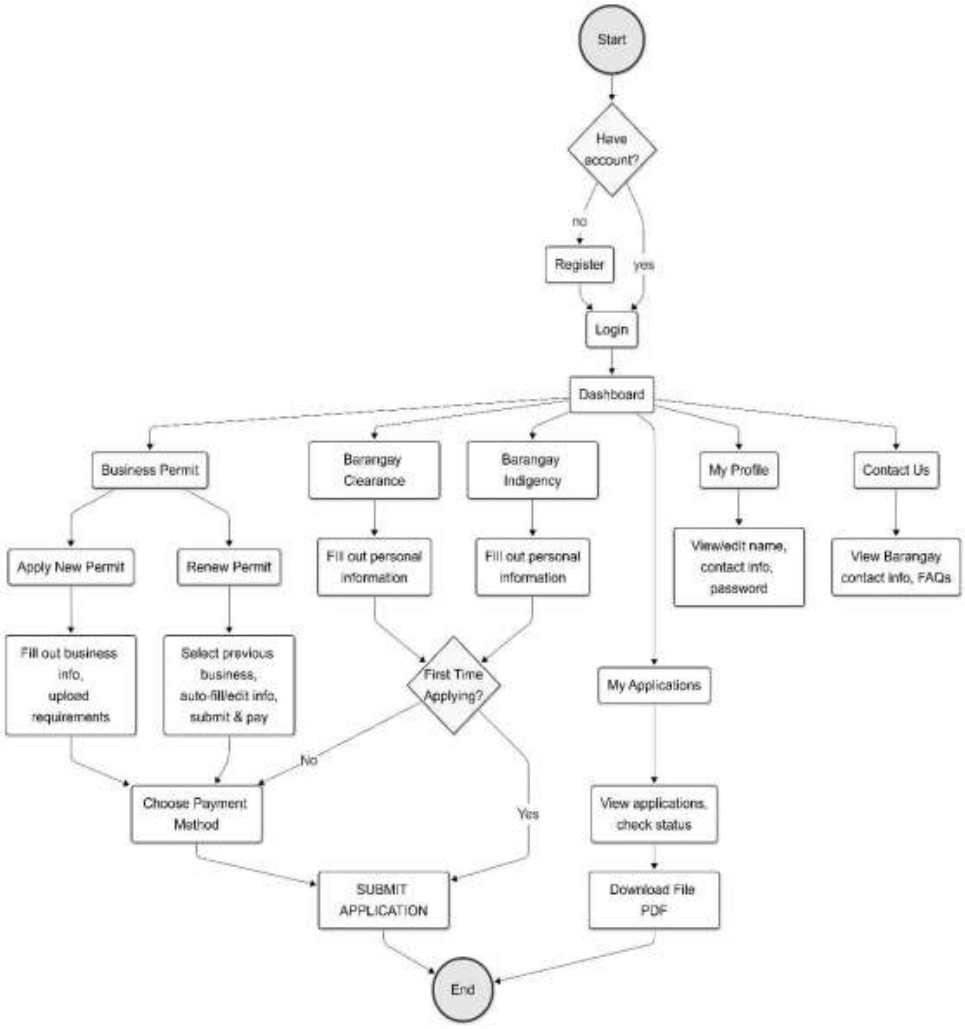
System Architecture

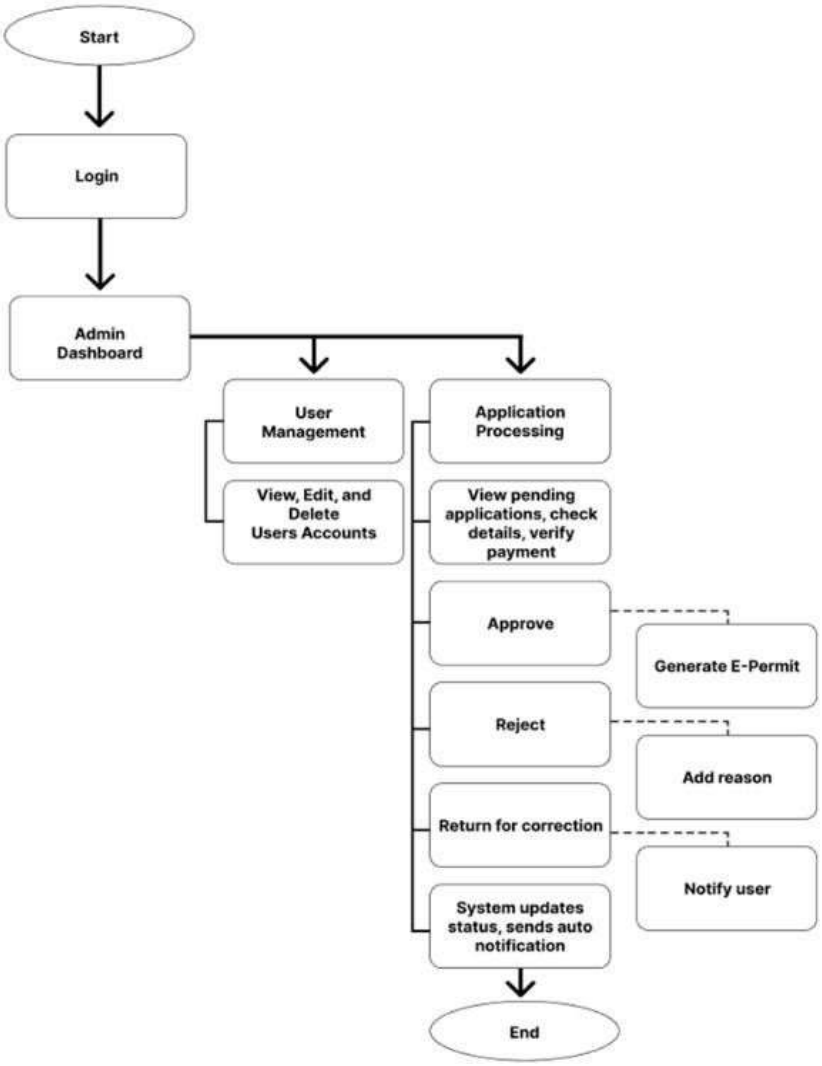


Block Diagram

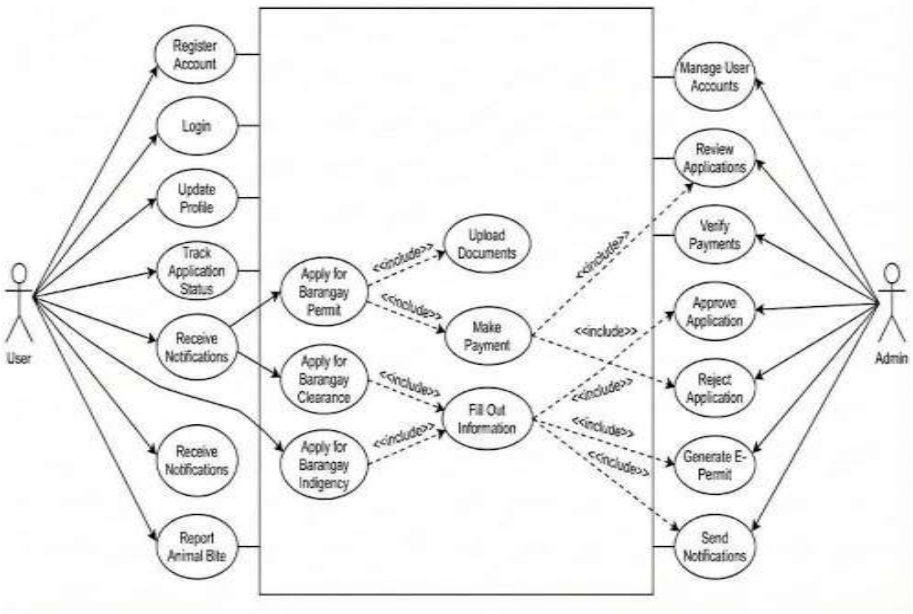


System Flowchart



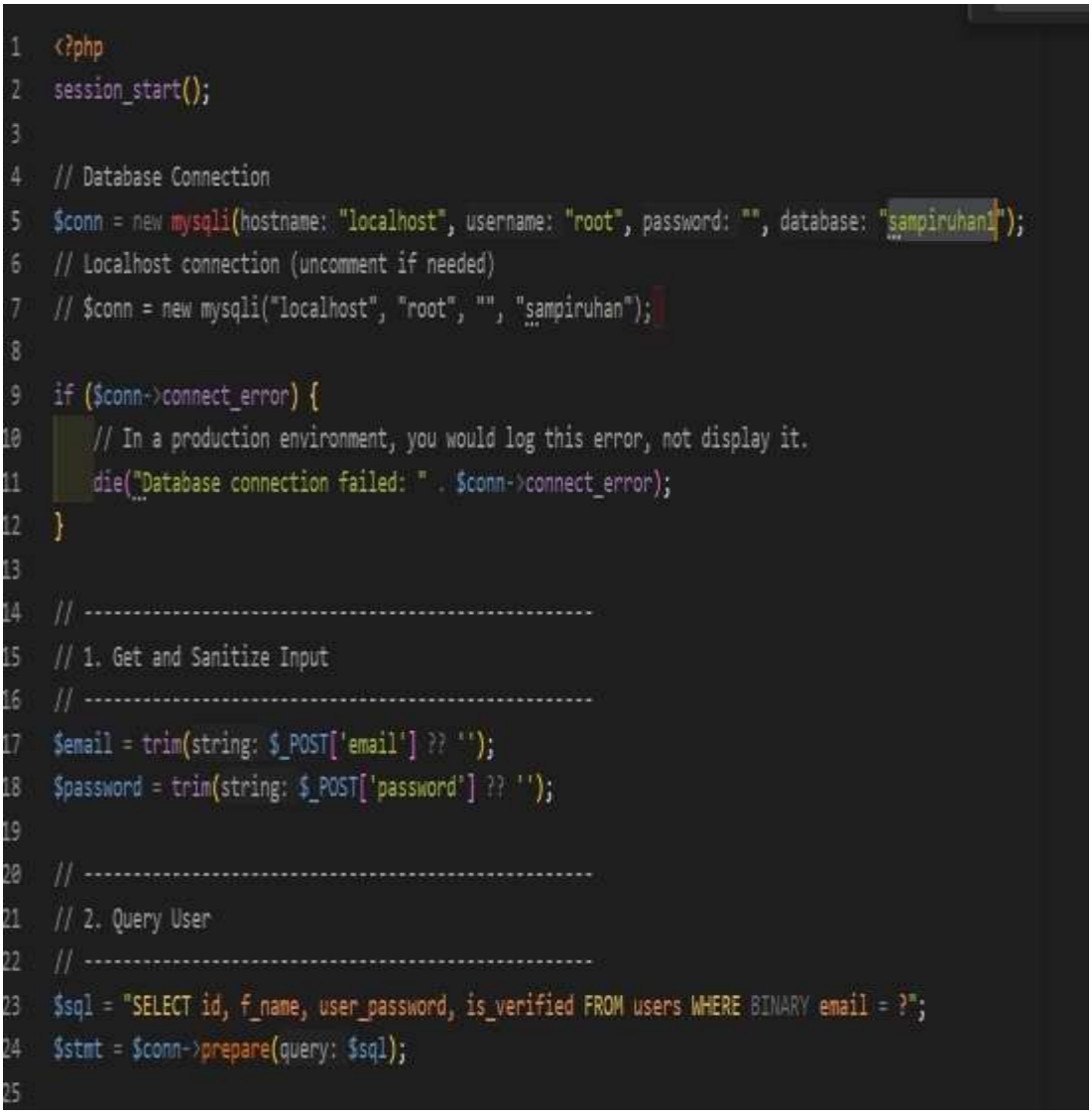


Used Case Diagram



APPENDIX C

Source Code



```php
<?php
session_start();

// Database Connection
$conn = new mysqli(hostname: "localhost", username: "root", password: "", database: "sampiruhan1");
// Localhost connection (uncomment if needed)
// $conn = new mysqli("localhost", "root", "", "sampiruhan");

if ($conn->connect_error) {
    // In a production environment, you would log this error, not display it.
    die("Database connection failed: " . $conn->connect_error);
}

// ------------------------------------------------
// 1. Get and Sanitize Input
// ------------------------------------------------
$email = trim(string: $_POST['email'] ?? '');
$password = trim(string: $_POST['password'] ?? '');

// ------------------------------------------------
// 2. Query User
// ------------------------------------------------
$sql = "SELECT id, f_name, user_password, is_verified FROM users WHERE BINARY email = ?";
$stmt = $conn->prepare(query: $sql);

```

Code Use in Login

APPENDIX

D

Communications Letter & Approvals

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

April 3, 2025

Honorable Roseller Rizal
Mayor, Calamba
Office of the Mayor
Complex, Bacnotan Road
Brgy. Real, City of Calamba, Laguna, 4027

Dear Mayor Rizal,

Greetings!

We hope this message finds you well. We are a group of students from **Laguna State Polytechnic University**, currently working on our capstone project titled **"Digital Document Management and Application System for Residents of Barangay Sampiruhan."**

This project aims to streamline the process of requesting and managing barangay documents through a digital platform. By minimizing manual paperwork and reducing processing time, the system seeks to make document transactions more efficient, transparent, and convenient for both residents and barangay staff.

We wanted to keep you informed of our work and would be deeply grateful for your support as we move forward with this initiative.

Thank you for your time and consideration.

Respectfully,

Claire Marie Enriquez
Project Leader
BSIT Student, Laguna State Polytechnic University

Mekhaila Eusebio
Team Member
BSIT Student, Laguna State Polytechnic University

Michaela Sta. Clara
Team Member
BSIT Student, Laguna State Polytechnic University

APPROVAL SHEET

The Capstone Project entitled **"Digital Document Management and Application System for Residents of Barangay Sampiruhan"** prepared and submitted by **Claire Marie A. Enriquez, Mekhaila S. Eusebio and Michaela D. Sta. Clara** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program**,** is hereby recommended for approval and acceptance.

JONALYN JOY B. LABAYNE
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

ROWAN ELOMINA
Member

CAROLINA R JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program at Laguna State Polytechnic University-Los Banos Campus.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, Ed. D.
Chairperson, Research and Development Services

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Dear IT Expert,

Good day! We are 4th year students conducting a research study and developing a system entitled, "**Digital Document Management and Application System for Residents of Barangay Sampiruhan".** We are currently pursuing a BS in Information Technology specializing in Service Management Program. We now seek to determine if the said system adheres to ISO 25010 standards. To assess the product quality, we have devised a rubric. Your responses will help us evaluate various aspects of the system based on the said international standard.

Sincerely,"
Claire Marie Enriquez, Mekhaila Eusebio, and Michaela D. Sta. Clara
LSPU Los Baños, College of Computer Studies
09359068956

ISO 25010 RUBRIC
Please encircle the score associated with each category

FUNCTIONAL SUITABILITY

Sub-Attributes	**Indicator**	**5 – Excellent**	**4 – Very Good**	**3 – Good**	**2 – Fair**	**1 – Poor**
Functional Completeness	The system provides all necessary functions for document application and approval.	All functions are complete and effective.	Nearly all functions are present and effective.	Most functions are available.	Some functions missing or incomplete.	Many functions unavailable or not working.

Functional Correctness	The system accurately processes and generates barangay documents.	Always accurate and correct outputs.	Minor, rare errors.	Occasional small errors.	Errors often affect output.	Frequent inaccuracies.
Functional Appropriateness	The features are relevant and beneficial for residents and staff.	Fully supports users' needs.	Supports most user needs.	Adequately relevant.	Limited usefulness.	Not suitable for user needs.

PERFORMANCE EFFICIENCY

Sub-Attributes	**Indicator**	**5 – Excellent**	**4 – Very Good**	**3 – Good**	**2 – Fair**	**1 – Poor**
Time Behavior	The system responds and loads quickly during use.	Very fast and responsive.	Usually fast; minimal delay.	Average response time.	Often slow.	Unresponsive or lags frequently.
Resource Utilization	The system efficiently uses network and device resources.	Highly optimized performance.	Efficient most of the time.	Acceptable usage.	Sometimes resource heavy.	Poorly optimized and wasteful.

COMPATIBILITY

Sub-Attributes	Indicator	5 – Excellent	4 – Very Good	3 – Good	2 – Fair	1 – Poor
Coexistence	The system works well with other barangay tools or platforms .	Fully compatible.	Mostly compatible.	Some issues occur occasionally .	Limited compatibility .	Not compatible .
Interoperability	The system exchanges data effectively with related applications.	Excellent data exchange and communication.	Good integration.	Partial interoperability.	Limited data exchange.	Cannot integrate with other systems.

USABILITY

Appropriateness Recognizability	The interface is clear and easy to understand .	Highly intuitive and well structured layout.	Clear layout; minor issues.	Generally easy to use.	Some confusing parts.	Unclear and disorganized .
Learnability	The system is easy to learn even for first time users.	Very easy; no help needed.	Easy with minimal help.	Understadable with effort.	Hard to learn.	Very difficult to learn.

Operability	Users can easily perform functions (submit, track, pay).	All functions are smooth and simple.	Most functions easy to perform.	Manageable but not seamless.	Some confusing steps.	Hard to operate.
User Error Protection	Prevents mistakes or guides users to correct actions.	Strong prevention and clear guidance.	Effective error handling and tips.	Some error handling.	Limited guidance.	No error protection or help.
User Engagement / Satisfaction	The system keeps users motivated to	Very engaging and satisfying.	Engaging.	Somewhat engaging.	Slightly engaging.	Not engaging at all.
	participate.					
User Interface Aesthetics / Helpfulness	Provides clear help, tooltips, or guidance.	Complete help tools and tips.	Helpful guidance.	Some assistance.	Minimal help.	No help or guidance.

REALIABILITY

Sub-Attributes	**Indicator**	**5 – Excellent**	**4 – Very Good**	**3 – Good**	**2 – Fair**	**1 – Poor**
Maturity	The system operates without errors or crashes.	Always stable.	Rare, minor bugs.	Occasionally unstable.	Frequent bugs.	Often crashes.

Availability	The system is consistently available for use.	Always accessible.	Almost always available.	Occasionally down.	Often down.	Rarely available.
Recoverability	Data and operations can be restored after errors.	Quick, full recovery.	Recovers well.	Partial recovery.	Slow recovery.	Cannot recover data.

SECURITY

Sub-Attributes	**Indicator**	**5 – Excellent**	**4 – Very Good**	**3 – Good**	**2 – Fair**	**1 – Poor**
Confidentiality	Protects user data from unauthorized access.	Strong protection.	Secure with few concerns.	Some vulnerability.	Risky protection.	Not secure.
Integrity	Ensures data remains accurate	Fully maintains accuracy.	Mostly consistent.	Minor inconsistencies.	Often inaccurate.	Easily corrupted.
	and unaltered.					
Authenticity	Verifies user and document authenticity.	Very reliable authentication.	Generally authentic.	Sometimes uncertain.	Weak verification.	No authentication.

Accountability	Tracks actions and changes by users/staff.	Full audit trail.	Good tracking.	Partial tracking.	Minimal logs.	No tracking at all.

MAINTANABILITY

Sub-Attributes	**Indicator**	**5 – Excellent**	**4 – Very Good**	**3 – Good**	**2 – Fair**	**1 – Poor**
Modularity	The system's components are well-organized for updates.	Very modular.	Mostly modular.	Moderate modularity.	Some dependenc y issues.	Poorly structured.
Reusability	Code/components can be reused.	Highly reusable.	Some reusable parts.	Moderate reuse.	Limited reuse.	Not reusable.
Analyzability	System issues can be easily analyzed.	Easy to diagnose.	Fairly easy.	Manageable.	Hard to analyze.	Very difficult to analyze.
Modifiability	The system can be easily modified or updated.	Very flexible and easy to modify.	Easy to modify.	Manageable with effort.	Hard to modify.	Not modifiable.
Testability	Easy to test for bugs and improvements.	Fully testable.	Easy to test.	Moderately testable.	Difficult to test.	Not testable.

PORTABILITY

Sub-Attributes	Indicator	5 – Excellent	4 – Very Good	3 – Good	2 – Fair	1 – Poor
Adaptability	Works on different devices (PC, phone, tablet).	Fully adaptive across all devices.	Works on most devices.	Adequate responsiveness.	Limited device support.	Not adaptive.
Scalability	Can support growing numbers of users.	Scales efficiently.	Handles growth well.	Moderate scalability.	Limited scaling.	Cannot handle growth.
Installability	Easy to deploy and install.	Very easy installation.	Simple installation process.	Manageable setup.	Complex installation.	Very difficult to install.
Replaceability	Can be replaced or upgraded without disruption.	Easily replaceable.	Replaceable with effort.	Some difficulty.	Hard to replace.	Not replaceable.

SAFETY

Sub-Attributes	Indicator	5 – Excellent	4 – Very Good	3 – Good	2 – Fair	1 – Poor

Operational Safety	Operates safely within expected parameters.	No risks at all.	Very few minor risks.	Occasionally risky.	Often risky.	Unsafe to operate.
Risk Detection and Prevention	Identifies and prevents risks.	Fully detects and prevents risks.	Detects most issues.	Sometimes misses risks.	Often fails to detect.	No detection or prevention.
Data and System Protection	Prevents data loss or malfunction.	Fully protected and reliable.	Mostly safe and stable.	Sometimes fails.	Often fails.	No protection at all.
Issue Alert System	Provides alerts for issues or risks.	Always sends timely alerts.	Usually alerts users.	Occasional delay in alerts.	Rare alerts.	No warning system.
Secure Integration	Integrates securely with other systems.	Seamless and safe integration.	Mostly secure.	Some issues during integration.	Risky connection.	Unsafe integration.

_____ _____

Signature over printed name
Survey Questionnaire Validator

Date Signed

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

Dear Respondent,

Good day! We are 4th year students pursuing a BS in Information Technology specializing in Service

Management Program conducting a research study and developing a system entitled, "**Digital Document Management and Application System for Residents of Barangay Sampiruhan".** We seek to determine the acceptability of the technology developed by asking for your opinion using the indicators below. Please use the description for each score so that you will be guided accordingly. Rest assured that any information you provide will be treated with the utmost confidentiality in accordance with RA 10173 (Data Privacy Act) and will be used solely for research purposes. Thank you for your cooperation.

Sincerely,

Claire Marie Enriquez, Mekhaila Eusebio and Michaela D. Sta. Clara
LSPU Los Baños, College of Computer Studies 09359068956

Technology Acceptance Model Rubric

Please encircle the score associated with each category

Name: _____
Age: ____

1. Perceived Usefulness. Measures how well the system improves productivity, efficiency, and overall usefulness in obtaining documents and making payments.

Indicator	4 - Strongly Agree	3 - Agree	2 - Disagree	1 - Strongly Disagree
1. The system helps in quickly obtaining documents (e.g., Clearance, Certification).	Significantly reduces time and effort in document acquisition.	Facilitates and speeds up the process.	Improves slightly; some delays still occur.	No change; takes as long as the traditional process.
2. The digital payment option (e.g., GCash) offers greater convenience.	Provides maximum convenience and I prefer it over cash.	Facilitates the process and is a good option.	Helped slightly; I still prefer cash.	Does not add to convenience.

3. Does not add to convenience.	Provides excellent clarity on application status; no need to inquire at the office.	Provides sufficient status information.	Status information is limited.	Does not display a clear status.
4. The notification system (email/SMS) is useful for knowing if the application is approved or declined.	Effectively provides reminders and immediate application results.	Helps update the application status.	Provides minimal help and is not always timely.	The notification has no benefit.
5. The system improves the quality of service provided by Barangay Sampiruhan. of community engagement.	Greatly enhances the overall service and image of the barangay.	Improves some aspects of the service.	Provides minimal help but is not very significant.	Has no effect on service quality.

2. Perceived Ease of Use. Measures simplicity, usability, and learning comfort of the system.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. The interface (design) of the web system is easy to understand and use.	Intuitive and very clear; everything needed is easy to find.	Easy to use but with a few minor confusions.	Sometimes difficult to use and hard to know where to start.	Complicated and confusing interface.
2. The process for applying for a document (e.g., Clearance) is simple and straightforward.	The application steps are very simple and clear.	Simple but with a few unclear steps.	A bit complicated or timeconsuming to complete.	Confusing and hard to complete.
3. Learning the system is quick and effortless.	3. Learning the system is quick and effortless.	Takes little time to learn.	Requires repeated trials to understand.	Very hard to learn or remember.
4. Selecting and using the payment options (cash/GCash) is clear and easy.	The payment process is fully clear and transparent.	Mostly clear but slightly complex.	Requires extra explanation.	Hard to understand and confusing.

5. The system can be used without technical assistance.	Completely selfexplanatory and simple to use without any help.	Usable with minimal assistance.	Requires guidance or instructions.	Difficult to use without constant help.

3. Behavioral Intention to Actual Use. Measures the user's intention to continue using and integrate the system into regular practice.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. Intention to continue using the system for document applications.	Strongly intends continued use on all occasions.	Likely to continue using.	Unsure about continued use.	No intention to use again.
2. Intention to recommend the system to other barangay residents.	Actively recommends the system to others.	Sometimes recommends.	Rarely recommends.	Does not recommend.
3. Preference for this digital system over the traditional process of going to the office.	Fully prefers this system over traditional methods.	Prefers but still uses traditional methods.	Neutral or indifferent between both.	Prefers traditional application at the office only.
4. Intention to use the system frequently whenever a document is needed.	Frequently uses the system.	Occasionally uses the system.	Rarely uses the system.	Rarely uses the system.
5. Intention to utilize the online payment feature (GCash) for all transactions.	Strongly intends to use the online payment feature.	Likely to use the online payment.	Unsure or will only use cash.	No intention to use online payment.

4. Attitude Toward Using the System. Measures user satisfaction, comfort, and emotional response to using the system.

Indicator	4 - Excellent	3 - Good	2 - Fair	1 - Poor
1. Enjoyment while using the system.	Greatly enjoys using it.	Enjoys moderately.	Rarely enjoys.	Does not enjoy at all.

2. The system builds confidence in the barangay's service.	Strongly believes it every confident and comfortable with the barangay's service because of enhances participation.	Confidence increases somewhat.	Slightly uneasy.	Not confident or comfortable at all.
3.The system provide positive experience applying documents.	Very positive overall impression.	Generally positive impression.	Neutral or mixed feelings.	Negative impression.
4. Feeling that document is acquisition and more modern effective because of th system.	Strongly believes this is the more modern way.	Believes it helps somewhat.	Unsure about the impact.	Does not believe it is more effective.
5. The system builds confidence in the barangay's service.	Very confident and comfortable with the barangay's service because of it.	Confidence increases somewhat.	Slightly uneasy.	Not confident or comfortable at all.

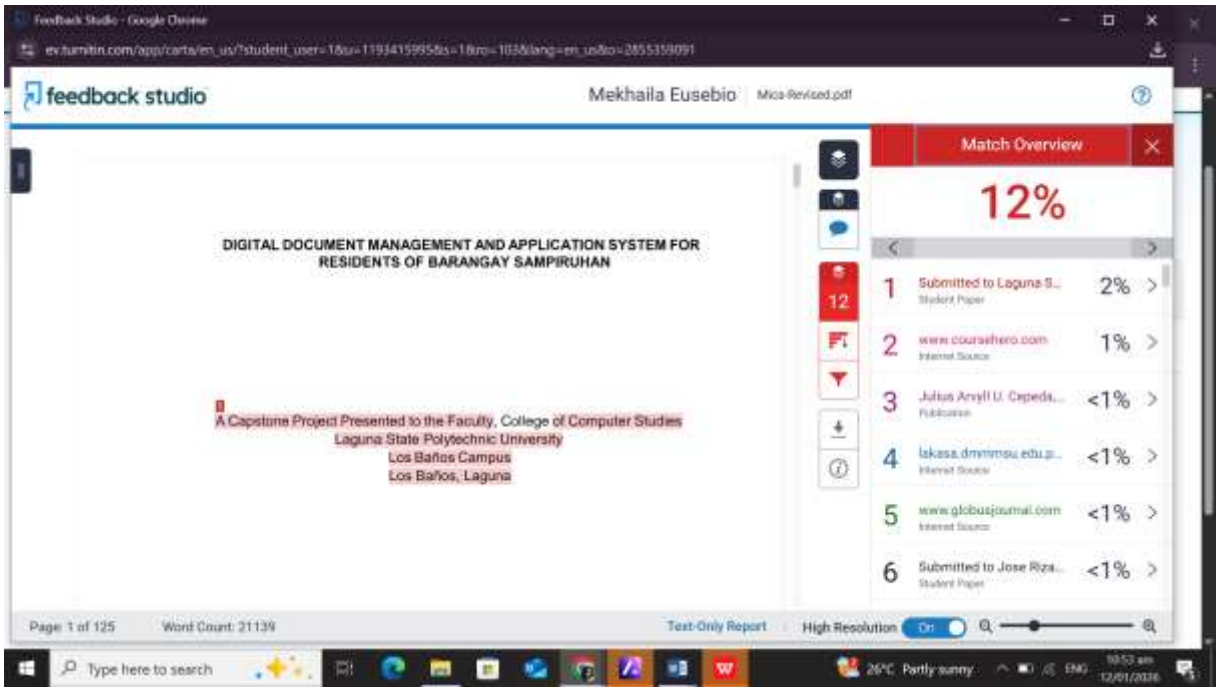
_____ _____

Signature over printed name Date Signed

Respondent

APPENDIX E

Plagiarism Result


feedback studio
Mekhaila Eusebio
Mica-Revised.pdf
Match Overview
12%
1 Submitted to Laguna S... 2%
2 www.coursehero.com 1%
3 Julius Arryll U. Cepeda... <1%
4 lakasa.dmmmsu.edu.p... <1%
5 www.globusjournal.com <1%
6 Submitted to Jose Riza... <1%
DIGITAL DOCUMENT MANAGEMENT AND APPLICATION SYSTEM FOR RESIDENTS OF BARANGAY SAMPIRUHAN
A Capstone Project Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna
Page: 1 of 125
Word Count: 21139
Text-Only Report
High Resolution
On
Type here to search
26°C Partly sunny

APPENDIX F

Documentation










DEFENDED

APPENDIX G

Publishable Paper

Digital Document Management and Application System for Residents of Barangay Sampiruhan

Claire Marie A. Enriquez, Mekhaila S. Eusebio, Michaela D. Sta. Clara

College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

Article history:
Received by
Revised by
Accepted by

Keywords:

Digital Document Management
Web Application
Laravel
GIS
Public Service
Barangay Residents

ABSTRACT

This study introduces a web-based Digital Document Management and Application System designed to make document submission, processing, and tracking more efficient for the residents of Barangay Sampiruhan. The system enables residents to submit requests online while allowing barangay officials to manage and monitor applications with ease and accuracy. Built using Laravel and supported by Geographical Information System (GIS) mapping, the system ensures data is securely organized and accessible, minimizing manual errors and delays. Findings indicate that the platform improves communication between residents and officials, enhances transparency, and streamlines administrative workflows. By adopting this digital approach, Barangay Sampiruhan can provide faster, more reliable public service, while fostering a more connected and responsive community.

In addition to these core benefits, the study highlights how the system encourages greater community participation by making essential services more accessible to residents, especially those who face challenges visiting the barangay office in person. The digital platform also provides a centralized database where documents can be securely stored, retrieved, and managed, ensuring accuracy and consistency in record-keeping. Overall, the system demonstrates the potential of digital innovation in strengthening local governance and improving service delivery, serving as a model for future technology-driven initiatives in similar communities.

I. INTRODUCTION

Barangay Sampiruhan, situated in the City of Calamba, Laguna, is home to a vibrant community of 9,466 residents as of the 2020 Census. Spanning 81 hectares, the barangay encompasses residential, agricultural, and commercial areas, with a rich historical significance, notably as the site of the Doce de Pebrero Massacre during World War II. This event is commemorated by the Putol na Kandila monument, honoring the civilian victims of the massacre.

As urbanization accelerates, the need for efficient, accessible, and transparent public services has become more pressing. Currently, Barangay Sampiruhan relies on a manual system to process documents such as barangay clearances, business permits, and other administrative tasks. However, this traditional approach has proven inadequate, often leading to long wait times, delayed approvals, misplaced records, and a

heavy workload for barangay personnel. These inefficiencies hinder service delivery and compromise the overall administrative functioning of the barangay.

In response to these challenges, this study proposes the development of a Digital Document Management and Application System for residents of Barangay Sampiruhan. The web-based platform will streamline document processing, allowing residents to easily apply for and track barangay services, including clearances and certificates. One key feature of the system will be its integration with Republic Act No. 11261, ensuring that first-time job seekers receive free Barangay Clearances and Certificates of Indigency, as required by law.

Beyond improving service efficiency, the proposed system also offers a strategic solution to safeguard against risks such as natural disasters, particularly flooding, which can damage physical records.
Digitizing records will not only enhance the security and accessibility of important documents but also increase the barangay's resilience during emergencies.

Furthermore, with the growing population and number of businesses in the area, a more structured and efficient transaction management system is crucial. Automation of administrative tasks will reduce delays, minimize human error, and foster greater transparency in public services.

Despite the promising benefits, the implementation of this system faces challenges, including technical difficulties, inconsistent internet connectivity, and varying levels of digital literacy among residents. In addition, safeguarding sensitive personal and financial information requires robust cybersecurity measures to ensure data protection and public trust. Overcoming these hurdles through regular maintenance, training programs, and security protocols is key to successful implementation.

This study highlights that the Digital Document Management and Application System for Barangay Sampiruhan represent a transformative step toward modernizing local governance. By enhancing operational efficiency, protecting critical records, and improving public service delivery, the system will not only streamline administrative processes but also reinforce trust between the barangay and its residents. As a result, it will contribute to the continued growth and development of the community while empowering residents with more accessible and reliable services.

I. RELATED LITERATURE

The integration of information and communication technology (ICT) in local governance has been widely recognized as a key driver of transparency, efficiency, and improved public service delivery. Carpio (2020) emphasized that many barangays in the Philippines still rely on manual, paper-based systems, which often result in delays, errors, and inefficiencies in processing documents such as barangay clearances and permits.[1] E-governance initiatives address these issues by digitizing administrative processes, enabling faster transactions, improved accessibility, and more reliable service

delivery for residents (Bautista, 2015; Siar, 2016).[2] These findings support the need for a digital document management system in barangay operations.

Several studies highlight how digital platforms enhance operational efficiency in local government units.[3] Gallera and Salvador (2019) found that automating business permit and document processing significantly reduces processing time

Despite the benefits of digital governance, challenges in system adoption remain prevalent, particularly in rural and semiurban communities. The National ICT Confederation of the Philippines (2023) and the Philippine Statistics Authority (2022) identified poor internet connectivity, limited access to devices, financial constraints, and low digital literacy as major barriers to successful egovernance implementation. Melendez (2021) emphasized that these challenges can be mitigated through infrastructure development, training programs, and stakeholder engagement. These findings are relevant to the proposed Barangay Sampiruhan system, which considers user training and system sustainability.

Electronic payment systems have also been shown to improve efficiency and transparency in local government transactions.[7] Manlapig (2021) and Santiago et al. (2020) discussed how QR code–based and mobile payment platforms streamline payments, reduce cash handling, and create reliable audit trails.[8] Studies further indicate that platforms such as GCash and PayMaya are accessible even to unbanked populations, making them suitable for barangay-level services (Zoleta, 2021; Agur, Ari, & Catão, 2020).[9] These findings support the integration of digital payment features in the proposed system to improve transaction speed and accountability.[10]

Barangay Management Information Systems (BMIS) and Barangay Management Systems (BMS) provide established models for digitizing barangay operations.[11] Kazuya (2015) [4]while improving compliance and citizen satisfaction. Similarly, Nuvitek (2025) noted that automation minimizes manual workload and enhances accuracy.[5] The integration of AI-driven and automated systems further improves approval workflows and workload management while reducing human errors (Brightly, 2020; Forbes, 2025).[6] These studies validate the proposed system's focus on automating barangay document applications and approvals.[7]

demonstrated that BMIS with SMS notification capabilities improves communication between barangay officials and residents while enhancing record accuracy and accessibility.[12] Meanwhile, Mirasol (2015) showed that automated barangay systems comply with the Local Government Code by maintaining updated resident records and streamlining permit and certification issuance. These systems serve as foundational references for the proposed Digital Document Management and Application System for Barangay Sampiruhan, reinforcing its relevance and practicality.

III. METHODOLOGY

Project Design

Figure 1 presents the Barangay Document Management and Application System features a modular architecture that integrates user access, application processing, centralized data storage, and administrative control to improve the delivery of barangay services. The system allows residents to securely register, submit applications for permits and certifications, track request status, and download approved documents through a userfriendly interface. On the administrative side, barangay officials can efficiently verify applications, manage users, validate payments, and generate reports through a secured management module. By connecting front-end user services with backend administrative functions via a unified database, the system enhances operational efficiency, transparency, and accountability. Its scalable design also

allows for future expansion to accommodate additional services and evolving barangay needs.

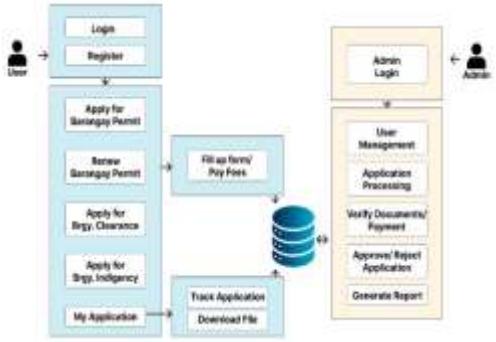


Figure 1. System Architecture **Project Development**

The system titled "Digital Document Management and Application System for Residents of Barangay Sampiruhan" will be developed using the Agile Scrum methodology. This iterative and flexible approach allows for continuous feedback and collaboration, ensuring the system meets the specific needs of residents and barangay officials.

The project will be organized into one- to two-week sprints, with regular stand-up meetings to monitor progress, address issues, and incorporate stakeholder feedback. Key components such as online document submission, real-time request tracking, dashboard management, and secure data handling will be developed and integrated incrementally. This approach enables rapid, adaptive development focused on improving efficiency, transparency, and accessibility in managing barangay documents.



Figure 2. Agile Scrum Methodology

IV. RESULTS AND DISCUSSION

Results of ISO 25010 and Technology and Acceptance Model.

Table 1 presents the results of the survey answered by the IT expert, showing the system's evaluation based on the ISO 25010 software quality model. The overall mean score of 4.26 (Excellent) indicates that the developed system met the expected software quality standards and was highly acceptable to the IT expert.

The system obtained Excellent ratings in most characteristics such as Function Suitability (4.33), Usability (4.33), Security (4.42), Maintainability (4.27), Portability (4.58), and Safety (4.60), demonstrating that the system performs effectively, is user-friendly, secure, and adaptable to various environments. Meanwhile, Performance Efficiency (3.67), Compatibility (4.00), and Reliability (4.11) were rated Very Good, signifying that while there is room for optimization, the system remains efficient and dependable. The evaluation result shows that the developed system achieved an Excellent level of acceptability and quality, satisfying most of the ISO 25010 characteristics evaluated by the IT expert.

Table 1. Results of survey answered by the IT expert

Indicators	Mean	SD	Interpretation
Perceived Usefulness	3.52	0.18	Strongly Agree
Perceived Ease of Use	3.48	0.16	Strongly Agree
Behavioral Intention to Use	3.51	0.17	Strongly Agree
Attitude Toward Use	3.55	0.15	Strongly Agree
Overall Mean	3.52	0.17	Strongly Agree

Legend: 1.0-1.75: Strongly Disagree; 1.762.50: Disagree; 2.51-3.25 Agree; 3.26-4.0: Strongly Agree

Table 2 summarize the results indicate that respondents strongly agree with all evaluated indicators of the system. Perceived Usefulness obtained a high mean score, suggesting that users believe the system will significantly improve efficiency and effectiveness in barangay transactions. Perceived Ease of Use also achieved a strong agreement rating, indicating that respondents find the system easy to learn and operate. The high mean for Behavioral Intention to Use reflects respondents' willingness to adopt and continuously use the system once implemented. Similarly, Attitude Toward Use recorded the highest mean among all indicators, demonstrating a very positive perception and acceptance of the proposed system. Overall, the findings confirm that the Barangay Document Management and Application System is highly acceptable, user-friendly, and likely to be successfully adopted by residents and barangay personnel.

Table 2. Results of survey answered by the Respondents

Characteristics	Mean	SD	Interpretation
Function Suitability	4.33	0.77	Excellent
Performance Efficiency	3.67	0.58	Very Good
Compatibility	4.00	0.58	Very Good
Usability	4.33	0.75	Excellent
Reliability	4.11	0.53	Very Good
Security	4.42	0.58	Excellent
Maintainability	4.27	0.35	Excellent
Portability	4.58	0.58	Excellent
Safety	4.6	0.58	Excellent
Overall	4.26	0.12	Excellent

Legend: 5.00 - 4.20 Excellent, 4.19 – 3.40 Good, 3.39 – 2.60 Fair, 2.59 – 1.80 Need Improvement, 1.79 – 1.00 poor

V. CONCLUSION, SUMMARY AND RECOMMENDATION

This chapter presents a consolidated discussion of the study's results derived from the data analysis. It interprets the findings in relation to the research objectives and the identified problems outlined in the earlier chapters. The implications of the results are discussed to highlight the contribution of the study to the intended users and stakeholders.

Based on the findings, relevant conclusions are drawn and practical recommendations are proposed to improve the system and guide future researchers in conducting related studies or further system enhancements.

Summary

The Digital Document Management and Application System for Residents of Barangay Sampiruhan aimed to design and implement a web-based platform that improves the efficiency, accessibility, and transparency of barangay document processing. The study addressed the limitations of the traditional manual system, which often resulted in long processing times, misplaced records, and inconvenience for both residents and barangay personnel.

The researchers developed a webbased system that allows residents to submit applications for barangay documents such as clearances, certificates of indigency, and business permits online.
The system features secure user authentication, online document submission, real-time application tracking, automated email notifications, and integrated digital payment options. On the administrative side, barangay officials are provided with a centralized dashboard to review applications, verify documents and payments, approve or reject requests, and generate reports for monitoring and decision-making.

The system was implemented and evaluated in Barangay Sampiruhan, Calamba City, Laguna. Its effectiveness and quality were assessed using the ISO/IEC 25010 software quality standards and the Technology Acceptance Model (TAM). Evaluation results from IT experts, barangay staff, and residents indicated excellent ratings in functionality, usability, security, maintainability, and overall system performance. Respondents strongly agreed that the system is useful, easy to use, and preferable to the traditional manual process. Overall, the findings demonstrated that the Digital Document Management and Application System significantly enhance service delivery by reducing processing time, minimizing

manual errors, improving transparency, and increasing resident satisfaction. The system shows strong potential as a sustainable digital solution for barangay-level governance and may serve as a model for other local government units seeking to modernize public service operations.

Conclusion

Development of an Email-Based Notification System for Application Status Updates

This study successfully developed an email-based notification system that keeps residents informed about the status of their barangay document applications. The system automatically notifies applicants whether their .requests are approved or declined, ensuring clear, timely, and efficient communication between the barangay and residents. This feature addressed delays and uncertainty commonly experienced in manual processing and fulfilled the objective of improving communication efficiency.

Development of a Simple and UserFriendly Web-Based Application, Tracking, and Payment System

The study successfully created a simple and user-friendly web-based system that allows residents to submit and track applications for barangay documents, including clearances and certificates, entirely online. The system provided convenient payment options through cash or GCash. In compliance with Republic Act No. 11261, first-time applicants for Barangay Clearance or Certificate of Indigency were automatically exempted from payment. This objective was achieved by improving accessibility, transparency, and convenience in barangay document processing.

Evaluation of System Effectiveness and Usability Using TAM and ISO/IEC 25010 Standards

The effectiveness and usability of the developed system were evaluated using the Technology Acceptance Model (TAM) and ISO/IEC 25010 software quality standards. The evaluation results indicated high levels of perceived usefulness, ease of use, and user acceptance, as well as strong performance in functionality, reliability, security, and maintainability. These findings confirm that the system meets high-quality standards and is acceptable for actual use in barangay operations.

Recommendations

The Digital Document Management and Application System for Residents of Barangay Sampiruhan was able to meet its objectives and function effectively based on the results and evaluations conducted. However, further enhancements and improvements can be made to maximize its accessibility, efficiency, and user satisfaction. The following recommendations are suggested:

Develop a mobile application to enable residents to submit requests and monitor applications directly from their smartphones, increasing convenience and system adoption.

Implement a real-time chat feature to allow residents to communicate instantly with barangay staff, improving support and streamlining inquiries.

Integrate card payment options such as Visa or Mastercard to offer faster, more secure, and convenient payment methods.

Use QR code verification for document requests so that residents can present a unique code at the barangay office, allowing staff to quickly access request details and reduce errors.

Add SMS notifications to inform residents about application status, approvals, or required actions, ensuring timely communication and minimizing unnecessary follow-ups.

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APPENDIX H

Curriculum Vitae



SKILLS

- Quality Assurance & Quality Control
- Documentation and Reporting
- Analytical and Technical Skills
- Team Collaboration
- Computer Literacy (MS Office, Excel)
- Adaptable and Reliable
- Continuous learning mindset

ENRIQUEZ, CLAIRE MARIE A.

Andrea Paz, Barangay Barandal, Calamba City, Laguna
0945-9711-004
enriquezclaire311@gmail.com

OBJECTIVE

To obtain a position in a company that will utilize my knowledge and skills and contribute to the development of the company. To engage in new challenges and learn experiences that will promote personal and group growth.

EXPERIENCE

Beltontotoku Philippines, Inc
Quality Assurance / Quality Control (QA/QC)
December 1, 2020 - March 4, 2022, Lot 7, Phase 1, Carmelray Industrial Park 2, Brgy. Milagrosa, Calamba City, Laguna, Philippines.

- Conducted inspection of products to ensure compliance with company standards.
- Assisted in monitoring production processes and quality checks.
- Documented and reported non-conformity and corrective actions.
- Worked closely with production teams to maintain consistent quality output.

EDUCATION

Laguna State Polytechnic University – Los Baños (LSPU-LB)
Bachelor of Science in Information Technology (BSIT)
Service Management Program
Brgy. Malinta, Los Baños

Punta Integrated School
Junior - Senior High School
2019 - 2020, Barangay Punta, Calamba City, Laguna

Barandal Elementary School
2013 - 2014, Barangay Barandal, Calamba City, Laguna



SKILLS

- Data Entry and Document Processing
- Attention to Detail
- Time Management Skills
- Can work under pressure
- Positive Attitude

EUSEBIO, MEKHAILA S.

12820 Antipas Compound, Mayondon Los Baños Laguna
0906 - 2807- 956
kylaeusebio2002@gmail.com

OBJECTIVE

To obtain a challenging position that will enhance and develop all my skills and knowledge in that will contribute to the progress of this company.

EXPERIENCE

Greensect Nature Foods
Administrative Aide
January 7, 2022 - April 13, 2022, Garnet St. Umali Subd., Batong Malake, Los Banos Laguna

- Organized, filed, and maintained records, documents, and reports.
- Assisted with data entry, typing, scanning, and photocopying of documents.
- Ensured office supplies were monitored and properly stored.

EDUCATION

Laguna State Polytechnic University – Los Baños (LSPU-LB)
Bachelor of Science in Information Technology (BSIT)
Service Management Program
Brgy. Malinta, Los Baños

Colegio De Los Baños Laguna
Senior High School
2019 - 2021, Lopez Avenue, Los Baños Laguna

Los Baños Integrated School
Junior High School
2015-2019, Tagpuan Bayog, Los Baños Laguna

Mayondon Elementary School
2009 - 2015, Barangay Mayondon, Los Baños Laguna



STA. CLARA, MICHAELA D.

md.staclara0320@gmail.com
4 - Service Management Program
09993572512
356, Sampiruhan, Calamba City, Laguna

SKILLS

- Supervisor
- Barista
- Excellent Communication Skills
- Computer skills
- Service and Support
- Hardworking and can multitask

EDUCATION

Laguna State Polytechnic University
SERVICE MANAGEMENT PROGRAM
MALINTA, LOS BANOS

AMA COMPUTER COLLEGE
INFORMATION AND COMMUNICATION TECHNOLOGY
PARIAN, CALAMBA

Calamba Intergrated School
PALINGON, CALAMBA

Sampiruhan Elementary Shool
SAMPIRUHAN, CALAMBA

OBJECTIVE

To improve my knowledge and develop my skills to grow personally and professionally.

EXPERIENCE

ADOBO CONNECTION

SERVICE CREW - 9 MONTHS

- Assisted customers promptly and politely, ensuring smooth and friendly service at the counter
- Contributed to maintaining high customer satisfaction, consistently achieving a 100% positive feedback rating
- Suggested improvements to workflow and store processes to help enhance efficiency and service quality
- Recognized as Top Employee of the Month three times within a nine-month period for outstanding performance and dedication

CAFE TRIBU CALAMBA

BARISTA - 3 AND HALF YEARS

- Assisted customers with their orders and questions efficiently and with a welcoming attitude
- Maintained a high customer satisfaction score of 100% by delivering excellent service and well-crafted beverages
- Contributed ideas to streamline store operations and enhance service quality
- Awarded Best Employee of the Month during the first month of employment for exceptional performance
- Promoted to Supervisor in recognition of strong leadership skills, reliability, and consistent high performance

CAFE TRIBU CALAMBA

SUPERVISOR- PRESENT

- Delivered consistently high standards of customer service, achieving a 100% customer satisfaction rating
- Promoted to a supervisory position and demonstrated sustained excellence in operational oversight and issue resolution
- Served as a key strategic contributor to team planning and problem-solving
- Recognized for exemplary leadership as Leadership Excellence Award of the Year recipient

TRAININGS

- Emersion in AMA COMPUTER COLLEGE admin staff - 2019